# Clinician routing knowledge base

Roles, scenarios and the patient facts each one looks for

| | |
|---|---|
| **Status** | DRAFT v0.1 (29 October 2026). For clinician review. Not clinical advice. |
| **Purpose** | Route a deterministic engine: free text -> role and bundle -> list of fact slots -> search in the patient documents -> sourced answer. Same input always gives the same slot list. |
| **What this file is not** | It holds no patient facts. Brand names and German words are search aids only; nothing here is shown to a clinician as a patient fact. |
| **Scale** | 20 roles, 18 scenario bundles, 63 fact slots, 22 link rules, 10 consistency checks, 21 guideline entries, 5 medication label entries from 12 label sources. |
| **Companion file** | routing_knowledge_base.json holds the same content in machine-readable form; this PDF is the review copy. |
| **Evidence marks** | V = guideline located and its title, date and the points listed were seen in search results on 29 Oct 2026 (section numbers still need a check against the full document). M = not confirmed; stays off the screen. Nothing here has been clinically reviewed yet. |

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# 1 How the engine uses this file, and conventions

## Flow

| Step | Name | What happens |
|---|---|---|
| 1 | Parse | One constrained LLM call turns the question into {role_id, bundle_ids, triggers} using only IDs from this file. Output is cached by normalised question text. Fuzzy alias matching (sections 2 and 4) is the cross-check; if both fail, bundle B18 is used with a visible note. |
| 2 | Expand | Bundle -> slot list with priorities. Role modifiers add or promote slots. 'implies' adds bundles (B04 adds B05). Union, then sort by priority, then by fixed slot order, so layout is identical every time. |
| 3 | Look up | Each slot is read from the pre-extracted, validated fact store built offline. Each result is found (with linked passage), documented as none, or not documented. |
| 4 | Rules | Link rules (section 7) add facts whose trigger and fact conditions hold. Consistency checks (section 8) attach contradiction cards to the slots they touch. |
| 5 | Detail level | Short = P1 only. Long = P1 + P2 + P3 plus the 'why this is here' line and guideline badge. |
| 6 | Render | One screen: facts with source links, 'not documented' rows, contradiction cards, guideline badges kept visually separate from patient facts. |

## Conventions

| Item | Rule |
|---|---|
| Priority P1 | Must see. Shown in short mode. Anything safety-critical (allergies) is P1 in every bundle that can lead to a prescription or procedure. |
| Priority P2 / P3 | Context / on request. Shown in long mode only. |
| Time anchor | Time anchor: 'now' = date of the latest document in the record (for the demo patient: 15 April 2016). Present-tense questions mean 'as of the anchor'. 'Latest' means newest dated value, chosen in code. |
| Staleness | Rule R21: if the newest value is more than 12 months older than the anchor, show 'as of <month year>'. |
| Absence | 'Documented as none' (for example 'keine bekannten Allergien') is a finding with a source. 'Not documented' means nothing was found after searching all documents; the screen says what was searched. |
| Status | Use IBM's vocabulary on every fact: diagnoses positive, negative (ruled out), suspected, statuspost; procedures positive, statuspost, recommended (section 12). Never show suspected, ruled-out or recommended items as confirmed or done. Older medication lists do not override newer ones. |
| Date precision | Source dates are month precision (only 9 of 62 procedure dates have a day). Store precision on every fact; order inside a month by document date; show month and year. |
| Age | Compute age from the date of birth at the anchor date. patient.json holds today's age; never use it. |
| Canonical text | One canonical text per document: entities decoded, tags and <BR> handled, whitespace collapsed. Spans and quotes are computed on it; IBM's matched text is never used as a quote. |
| Drug schedules | Four numbers (morning, midday, evening, night) are shown as three, for example 1 0 0 0 -> 1-0-0 'once in the morning'. |
| Empty lab tables | Both discharge letters end with an empty lab table and only HbA1c is documented, so lab slots show 'not documented' with the empty table as the searched passage. |
| Guideline badge | Shown as 'why this is shown', separate from patient facts, with the guideline year. It is never presented as a patient fact. |
| No invented facts | Only quotes that exist verbatim in the cited passage are shown. Translations, generic drug names and calculated values (with inputs linked) are allowed. |

## Fact shapes (how each slot is displayed)

| Shape | Display |
|---|---|
| current_list | Table of items with dose, schedule, status and as-of date. |
| latest_value | Newest dated value with unit and date; adds 'no later value documented' when true. |
| trend | All dated values, then a calculated change (code, inputs linked). |

| Shape | Display |
|---|---|
| event_history | Earlier event with date, description and documented cause or outcome. |
| status_check | Something was recommended; show the evidence it happened, else 'not documented'. |
| checklist | Sub-items, each independently: found / documented as none / not documented. |
| plan | Planned item with date and who decided it. |
| laterality_finding | Finding with side, degree or size, method and date. |
| list_all | Dated list of everything of this kind. |

# 2 Role directory

Grouped by working context. 'Match words' feed the alias matcher; the default bundles are what the role usually needs. For roles with several bundles (for example GP) the task words in section 4 pick the bundle.

| Role | Group | Match words (EN) | Match words (DE) | Default bundles |
|---|---|---|---|---|
| Anaesthetist | Perioperative and procedural | anaesthetist, anesthetist, anaesthesiologist, anaesthetic registrar, anaesthetics | Anästhesist, Anästhesistin, Anästhesie, Narkosearzt | B01, B05 |
| Surgeon (breast / general) | Perioperative and procedural | surgeon, operating surgeon, breast surgeon, surgical registrar | Chirurg, Chirurgin, Operateur, Brustchirurg | B02 |
| Radiation oncologist | Perioperative and procedural | radiation oncologist, radiotherapist, clinical oncologist (RT) | Strahlentherapeut, Radioonkologe | B12 |
| Medical oncologist | Perioperative and procedural | oncologist, medical oncologist, cancer doctor | Onkologe, Onkologin, internistische Onkologie | B11 |
| Vascular surgeon | Perioperative and procedural | vascular surgeon, vascular consultant | Gefäßchirurg, Gefäßchirurgin | B17 |
| Pre-assessment nurse | Perioperative and procedural | pre-assessment nurse, pre-op nurse, preadmission nurse | Prämedikationsschwester, Pflege Prämedikation | B01 |
| Emergency physician | Acute care | emergency physician, ED doctor, A&E doctor, ER doctor, night duty, casualty | Notfallmediziner, Notaufnahme, Ambulanzarzt, Nachtdienst | B04, B05 |
| Paramedic / triage nurse | Acute care | paramedic, ambulance crew, triage nurse, EMT | Rettungssanitäter, Notfallsanitäter, Triage, Rettungsdienst | B16 |
| Neurologist / stroke unit | Acute care | neurologist, stroke physician, stroke unit, TIA clinic | Neurologe, Neurologin, Stroke Unit, TIA-Ambulanz | B04, B17 |
| Radiologist / radiographer | Acute care | radiologist, radiographer, CT radiographer, imaging | Radiologe, Radiologin, MTRA, Bildgebung | B05 |
| General practitioner | Longitudinal care | GP, general practitioner, family doctor, family physician, primary care | Hausarzt, Hausärztin, Allgemeinmediziner, praktischer Arzt | B03, B06, B09, B10, B07 |
| Diabetologist / endocrinologist | Longitudinal care | diabetologist, endocrinologist, diabetes specialist | Diabetologe, Endokrinologe, Diabetesambulanz | B03 |
| Cardiologist | Longitudinal care | cardiologist, heart specialist, cardiology | Kardiologe, Kardiologin, Herzambulanz | B09 |
| Pharmacist | Longitudinal care | pharmacist, clinical pharmacist, pharmacy | Apotheker, Apothekerin, klinische Pharmazie | B07 |
| Ophthalmologist | Longitudinal care | ophthalmologist, eye doctor, eye specialist | Augenarzt, Augenärztin, Ophthalmologe | B14 |
| Dentist | Longitudinal care | dentist, dental surgeon, oral surgeon | Zahnarzt, Zahnärztin, Kieferchirurg | B15 |
| Ward nurse | Allied health and nursing | nurse, ward nurse, staff nurse, discharge nurse | Pflegeperson, Diplomierte Gesundheits- und Krankenpflege, Stationsschwester | B08 |
| Physiotherapist | Allied health and nursing | physiotherapist, physio, rehab therapist | Physiotherapeut, Physiotherapeutin | B13 |
| Dietitian | Allied health and nursing | dietitian, dietician, nutritionist | Diätologe, Diätologin, Ernährungsberater | B10 |
| Admitting physician (medical) | Allied health and nursing | admitting doctor, medical registrar, doctor on call, house officer, admissions | Aufnahmearzt, diensthabender Arzt, Turnusarzt | B18 |

# 3 Role profiles

P1 slots are the union of the role's default bundles plus role modifiers. 'Often missing' lists facts the role expects but letters commonly lack; the screen shows these as 'not documented' instead of hiding them.

## Perioperative and procedural

### Anaesthetist

| | |
|---|---|
| **Typical situations** | Pre-operative visit; sedation for a procedure; regional or neuraxial planning; post-operative review. |
| **Mindset** | Is it safe to anaesthetise, and what has to be managed around the procedure: bleeding, allergy, cardiac reserve, airway, glucose. |
| **Default bundles** | B01 Pre-procedure check (anaesthesia); B05 Safety check before iodinated contrast |
| **Role modifiers** | Airway assessment (P1), Earlier anaesthesia and complications (P1), Lung disease and sleep apnoea (P2) |
| **P1 slots** | Antiplatelets and anticoagulants, Allergies and intolerances, with reaction, LVEF and echo findings, Planned procedure and date, Age and sex, Vascular imaging: duplex, CT/MR angiography, Glucose-lowering drugs, incl. insulin, Creatinine and eGFR, Airway assessment, Earlier anaesthesia and complications |
| **Often missing** | Kidney function, airway assessment, earlier anaesthesia problems, time of last blood-thinner dose. |
| **Guideline sources** | G01, G02, G03, G06, G07 |

### Surgeon (breast / general)

| | |
|---|---|
| **Typical situations** | Planning after tumour board; consent; post-operative complication review. |
| **Mindset** | Exactly what is operated on, which side, what stage and biology, and what could complicate surgery. |
| **Default bundles** | B02 Surgical planning after tumour board |
| **Role modifiers** | Antiplatelets and anticoagulants (P1), Allergies and intolerances, with reaction (P1) |
| **P1 slots** | Cancer type and histology, TNM stage and grade, Hormone receptors, HER2, Ki-67, Tumour side, quadrant and size, Tumour board recommendation, Planned procedure and date, Antiplatelets and anticoagulants, Allergies and intolerances, with reaction |
| **Often missing** | Axillary status, staging imaging, earlier cancer treatment. |
| **Guideline sources** | G11, G17 |

### Radiation oncologist

| | |
|---|---|
| **Typical situations** | IORT or external-beam planning; boost decisions. |
| **Mindset** | Where and how much to treat: side, site, stage, earlier radiotherapy, implanted devices. |
| **Default bundles** | B12 Radiotherapy planning (incl. intra-operative) |
| **Role modifiers** | none |
| **P1 slots** | Tumour side, quadrant and size, Planned procedure and date, Tumour board recommendation, TNM stage and grade, Earlier cancer treatment, Pacemaker or ICD |
| **Often missing** | Cardiac device status, earlier radiotherapy. |
| **Guideline sources** | G11, G17 |

### Medical oncologist

| | |
|---|---|
| **Typical situations** | Adjuvant therapy discussion; tumour board preparation. |
| **Mindset** | Tumour biology and fitness for systemic therapy: receptors, stage, heart, kidney, liver, blood counts. |
| **Default bundles** | B11 Systemic cancer therapy planning |
| **Role modifiers** | none |
| **P1 slots** | Hormone receptors, HER2, Ki-67, Cancer type and histology, TNM stage and grade, Age and sex, LVEF and echo findings, Planned procedure and date |
| **Often missing** | Kidney, liver and blood counts; performance status. |
| **Guideline sources** | G11, G17 |

### Vascular surgeon

| | |
|---|---|
| **Typical situations** | Carotid disease review; carotid endarterectomy consideration. |
| **Mindset** | Degree, side and symptoms of the stenosis, imaging, and antiplatelet cover. |
| **Default bundles** | B17 Carotid and vascular review |
| **Role modifiers** | Coronary disease, MI, stents, heart failure (P2), LVEF and echo findings (P2) |
| **P1 slots** | Carotid stenosis: side, degree, method, date, Stroke, TIA, amaurosis fugax, Vascular imaging: duplex, CT/MR angiography, Antiplatelets and anticoagulants, Lipid-lowering drugs |
| **Often missing** | Follow-up imaging, smoking history. |
| **Guideline sources** | G08, G17 |

### Pre-assessment nurse

| | |
|---|---|
| **Typical situations** | Nurse-led pre-operative screening. |
| **Mindset** | What the anaesthetic team will ask: allergies, drugs, function, home support. |
| **Default bundles** | B01 Pre-procedure check (anaesthesia) |
| **Role modifiers** | Mobility and functional capacity (P1), Living situation and support (P1) |
| **P1 slots** | Antiplatelets and anticoagulants, Allergies and intolerances, with reaction, LVEF and echo findings, Planned procedure and date, Age and sex, Mobility and functional capacity, Living situation and support |
| **Often missing** | Mobility, home support. |
| **Guideline sources** | G01, G02 |

## Acute care

### Emergency physician

| | |
|---|---|
| **Typical situations** | Neurological or visual symptoms; chest pain; allergic reaction; hypo- or hyperglycaemia. |
| **Mindset** | What could explain the presentation, what is dangerous to give, and what the patient already takes. |
| **Default bundles** | B04 Acute neurological or visual presentation (emergency); B05 Safety check before iodinated contrast |
| **Role modifiers** | Advance directive / resuscitation status (P2) |
| **P1 slots** | Earlier similar episodes and their cause, Stroke, TIA, amaurosis fugax, Carotid stenosis: side, degree, method, date, Antiplatelets and anticoagulants, Blood pressure drugs, Allergies and intolerances, with reaction, Vascular imaging: duplex, CT/MR angiography, Glucose-lowering drugs, incl. insulin, Creatinine and eGFR |
| **Often missing** | Kidney function, current glucose, time of last antithrombotic dose. |
| **Guideline sources** | G08, G12, G14, G17 |

### Paramedic / triage nurse

| | |
|---|---|
| **Typical situations** | Pre-hospital handover; triage. |
| **Mindset** | Fast safety handover: allergies, blood thinners, main diagnoses, medication, resuscitation wishes. |
| **Default bundles** | B16 Emergency or paramedic handover |
| **Role modifiers** | none |
| **P1 slots** | Allergies and intolerances, with reaction, Antiplatelets and anticoagulants, All diagnoses with dates and status, Current medication, all, Advance directive / resuscitation status, Age and sex |
| **Often missing** | Resuscitation status, social support. |
| **Guideline sources** | G18 |

### Neurologist / stroke unit

| | |
|---|---|
| **Typical situations** | TIA or stroke work-up; carotid symptom attribution. |
| **Mindset** | Prior events, vascular findings, antithrombotics, rhythm, risk factors and baseline deficits. |
| **Default bundles** | B04 Acute neurological or visual presentation (emergency); B17 Carotid and vascular review |
| **Role modifiers** | Neurological findings and deficits (P1), Head imaging, CT or MRI (P1), ECG and rhythm (P1) |
| **P1 slots** | Earlier similar episodes and their cause, Stroke, TIA, amaurosis fugax, Carotid stenosis: side, degree, method, date, Antiplatelets and anticoagulants, Blood pressure drugs, Allergies and intolerances, with reaction, Vascular imaging: duplex, CT/MR angiography, Lipid-lowering drugs, Neurological findings and deficits, Head imaging, CT or MRI, ECG and rhythm |
| **Often missing** | ECG or rhythm monitoring, neurological baseline, head imaging report. |
| **Guideline sources** | G08, G17 |

### Radiologist / radiographer

| | |
|---|---|
| **Typical situations** | Contrast CT or angiography request check. |
| **Mindset** | Contrast safety: reactions, kidney function, metformin, earlier imaging. |
| **Default bundles** | B05 Safety check before iodinated contrast |
| **Role modifiers** | none |
| **P1 slots** | Allergies and intolerances, with reaction, Vascular imaging: duplex, CT/MR angiography, Glucose-lowering drugs, incl. insulin, Creatinine and eGFR |
| **Often missing** | Kidney function. |
| **Guideline sources** | G12, G13, G14 |

## Longitudinal care

### General practitioner

| | |
|---|---|
| **Typical situations** | Diabetes review; hypertension review; open follow-ups; medication review; weight loss. |
| **Mindset** | Longitudinal control and loose ends: values over time, doses, and what was supposed to happen next. The task picks the bundle. |
| **Default bundles** | B03 Diabetes review; B06 Open follow-ups after a hospital stay; B09 Hypertension and cardiovascular review; B10 Weight change and nutrition review; B07 Medication reconciliation and review |
| **Role modifiers** | none |
| **P1 slots** | Glucose-lowering drugs, incl. insulin, HbA1c: value, unit, date, Weight, Diabetes: type and duration, Hospital stays, admission to discharge, Follow-ups recommended at discharge, Evidence that a follow-up took place, Carotid stenosis: side, degree, method, date, Blood pressure drugs, Blood pressure and heart rate, Hypertension and hypertensive crises, Height and BMI, All diagnoses with dates and status, Current medication, all, Stopped, paused or changed medication, Allergies and intolerances, with reaction |
| **Often missing** | Later values after discharge; kidney function; completion of follow-ups. |
| **Guideline sources** | G05, G09, G10, G15 |

### Diabetologist / endocrinologist

| | |
|---|---|
| **Typical situations** | Diabetes review; peri-operative diabetes plan. |
| **Mindset** | Glycaemic control, drug regimen, hypoglycaemia and complications. |
| **Default bundles** | B03 Diabetes review |
| **Role modifiers** | Hypoglycaemia (P1), Diabetic complications (P1), Stopped, paused or changed medication (P1) |
| **P1 slots** | Glucose-lowering drugs, incl. insulin, HbA1c: value, unit, date, Weight, Diabetes: type and duration, Hypoglycaemia, Diabetic complications, Stopped, paused or changed medication |
| **Often missing** | Hypoglycaemia history, complications screening. |
| **Guideline sources** | G05, G10 |

## Cardiologist

| | |
|---|---|
| **Typical situations** | Blood pressure and heart-function review; pre-operative cardiac question. |
| **Mindset** | Pump function, rhythm, blood pressure control, ischaemic disease, vascular risk. |
| **Default bundles** | B09 Hypertension and cardiovascular review |
| **Role modifiers** | LVEF and echo findings (P1), ECG and rhythm (P1), Coronary disease, MI, stents, heart failure (P1), Valve disease, prosthetic valves (P2) |
| **P1 slots** | Blood pressure drugs, Blood pressure and heart rate, Hypertension and hypertensive crises, LVEF and echo findings, ECG and rhythm, Coronary disease, MI, stents, heart failure |
| **Often missing** | ECG, lipids, kidney function. |
| **Guideline sources** | G07, G09 |

## Pharmacist

| | |
|---|---|
| **Typical situations** | Medication reconciliation; prescribing safety review. |
| **Mindset** | Accurate, complete list; doses; changes; allergies; kidney function; interactions. |
| **Default bundles** | B07 Medication reconciliation and review |
| **Role modifiers** | none |
| **P1 slots** | Current medication, all, Stopped, paused or changed medication, Allergies and intolerances, with reaction |
| **Often missing** | Kidney and liver function, stop dates. |
| **Guideline sources** | G15, G13 |

## Ophthalmologist

| | |
|---|---|
| **Typical situations** | Transient vision loss; diabetic or hypertensive eye review. |
| **Mindset** | Visual events, their side, vascular and diabetic causes, and eye follow-up. |
| **Default bundles** | B14 Eye or visual-event review |
| **Role modifiers** | none |
| **P1 slots** | Eye symptoms and eye follow-up, Stroke, TIA, amaurosis fugax, Carotid stenosis: side, degree, method, date |
| **Often missing** | Diabetic eye screening results. |
| **Guideline sources** | G05, G08 |

## Dentist

| | |
|---|---|
| **Typical situations** | Extraction or minor oral surgery. |
| **Mindset** | Bleeding risk, allergies, valve disease, diabetes control. |
| **Default bundles** | B15 Minor procedure and bleeding check (e.g. dental) |
| **Role modifiers** | none |
| **P1 slots** | Antiplatelets and anticoagulants, Allergies and intolerances, with reaction, Valve disease, prosthetic valves |
| **Often missing** | Coagulation results, valve status. |
| **Guideline sources** | G03 |

## Allied health and nursing

### Ward nurse

| | |
|---|---|
| **Typical situations** | Admission assessment; discharge handover. |
| **Mindset** | Safe admission and discharge: allergies, medication, mobility, follow-ups, support. |
| **Default bundles** | B08 Admission or discharge handover |
| **Role modifiers** | none |
| **P1 slots** | All diagnoses with dates and status, Current medication, all, Allergies and intolerances, with reaction, Follow-ups recommended at discharge, Mobility and functional capacity |
| **Often missing** | Mobility, social support, resuscitation status. |
| **Guideline sources** | G18 |

### Physiotherapist

| | |
|---|---|
| **Typical situations** | Post-operative rehab after breast surgery. |
| **Mindset** | Side and type of surgery, function, cardiac limits, bleeding risk. |
| **Default bundles** | B13 Post-operative rehabilitation |
| **Role modifiers** | none |
| **P1 slots** | Planned procedure and date, Tumour side, quadrant and size, Mobility and functional capacity |
| **Often missing** | Functional status, home situation. |
| **Guideline sources** | G18 |

### Dietitian

| | |
|---|---|
| **Typical situations** | Weight loss; diabetes diet advice. |
| **Mindset** | Weight trajectory, diagnoses, glucose-lowering drugs and appetite. |
| **Default bundles** | B10 Weight change and nutrition review |
| **Role modifiers** | HbA1c: value, unit, date (P1) |
| **P1 slots** | Weight, Height and BMI, All diagnoses with dates and status, HbA1c: value, unit, date |
| **Often missing** | Diet history, appetite, swallowing. |
| **Guideline sources** | G18 |

### Admitting physician (medical)

| | |
|---|---|
| **Typical situations** | Medical admission; unfamiliar presentation. |
| **Mindset** | Complete picture of diagnoses, drugs, allergies and recent stays. |
| **Default bundles** | B18 General core summary (fallback) |
| **Role modifiers** | Hospital stays, admission to discharge (P1), Antiplatelets and anticoagulants (P1) |
| **P1 slots** | Age and sex, All diagnoses with dates and status, Current medication, all, Allergies and intolerances, with reaction, Hospital stays, admission to discharge, Antiplatelets and anticoagulants |
| **Often missing** | Anything not in the letters; a visible note says so. |
| **Guideline sources** | G18 |

# 4 Scenario bundles

A bundle answers 'what does this task need', independently of who asks. Rows are in display order. Short mode shows P1; long mode shows all.

## B01 Pre-procedure check (anaesthesia)

Anaesthetic assessment before surgery, sedation or a procedure under anaesthesia.

| | |
|---|---|
| **Match words (EN)** | pre-operative, preop, pre-op, pre-assessment, anaesthesia, anaesthetic, sedation, before surgery, premedication, scheduled for |
| **Match words (DE)** | präoperativ, Prämedikation, Narkose, Anästhesie, Sedierung, vor der Operation, Narkosevisite |
| **Implies** | none |

| Slot | Priority | Shape | Why it is here |
|---|---|---|---|
| Antiplatelets and anticoagulants [med_antithrombotic] | P1 | current_list | Bleeding risk and choice of anaesthetic technique. Show drug, dose, schedule and as-of date. |
| Allergies and intolerances, with reaction [allergies] | P1 | checklist | Drug, latex and contrast reactions before anything is given. |
| LVEF and echo findings [lvef] | P1 | latest_value | Cardiac reserve before anaesthesia. Show value, date and the echo wording. |
| Planned procedure and date [planned_procedure] | P1 | plan | Anchors the task: procedure and date. |
| Age and sex [age_sex] | P1 | latest_value | Age and sex frame risk, dosing and screening. |
| Glucose-lowering drugs, incl. insulin [med_antidiabetic] | P2 | current_list | Glycaemic control and drug safety. |
| HbA1c: value, unit, date [hba1c] | P2 | latest_value | NICE NG45 asks whether HbA1c was tested in the last 3 months, so the date matters as much as the value. |
| Creatinine and eGFR [kidney_function] | P2 | latest_value | Drug dosing; NICE NG45 links kidney testing to comorbidity. Often not documented. |
| ECG and rhythm [ecg_rhythm] | P2 | latest_value | Rhythm and conduction baseline. |
| Weight [weight] | P2 | trend | Dosing, nutrition and disease trajectory. |
| Height and BMI [height_bmi] | P2 | latest_value | BMI for dosing, anaesthesia and metabolic risk. |
| Blood pressure and heart rate [vitals_bp_hr] | P2 | trend | Baseline haemodynamics. |
| Blood pressure drugs [med_antihypertensive] | P2 | current_list | Blood pressure control and interactions. |
| Haemoglobin and platelets [blood_count] | P2 | latest_value | Anaemia and platelet reserve. |
| Coagulation tests and bleeding history [coagulation] | P2 | latest_value | Bleeding risk. |
| Earlier anaesthesia and complications [prior_anaesthesia] | P2 | event_history | Earlier anaesthetic problems. |
| Airway assessment [airway] | P2 | latest_value | Airway planning. |
| Coronary disease, MI, stents, heart failure [coronary_hf] | P2 | list_all | Ischaemic and heart failure burden. |
| Carotid stenosis: side, degree, method, date [carotid_stenosis] | P2 | laterality_finding | Side and degree guide imaging and symptom attribution. |
| Lung disease and sleep apnoea [pulmonary_hx] | P2 | list_all | Respiratory reserve. |
| Mobility and functional capacity [mobility_function] | P2 | latest_value | Functional reserve and rehab needs. |
| All diagnoses with dates and status [diagnoses_all] | P2 | list_all | Comorbidity context. |
| Smoking and alcohol [smoking_alcohol] | P3 | latest_value | Perioperative and vascular risk modifiers. |
| All procedures and operations [procedures_all] | P3 | list_all | Earlier operations and interventions. |
| Valve disease, prosthetic valves [valves] | P3 | latest_value | Structural heart disease. |
| Pacemaker or ICD [cardiac_device] | P3 | latest_value | Devices affect procedures and radiotherapy. |

| Slot | Priority | Shape | Why it is here |
|---|---|---|---|
| Analgesics and psychotropic drugs [med_analgesic_psychotropic] | P3 | current_list | Sedation, interaction and bleeding relevance. |
| Stroke, TIA, amaurosis fugax [stroke_tia_hx] | P3 | event_history | Cerebrovascular events. |
| Current medication, all [med_current] | P3 | current_list | Complete current medication. |

| | |
|---|---|
| **Expected but may be missing** | Creatinine and eGFR, Airway assessment, Earlier anaesthesia and complications, Coagulation tests and bleeding history, ECG and rhythm, Height and BMI, Smoking and alcohol, Mobility and functional capacity |
| **Link rules** | R01, R06, R07, R08, R17, R18, R19, R21 |
| **Guideline sources** | G01, G02, G03, G06, G07, G16 |
| **Design note** | Task before role: the same bundle serves anaesthetist, pre-assessment nurse and, with modifiers, surgeon. |

## B02 Surgical planning after tumour board

Surgeon preparing an operation for a diagnosed cancer after the multidisciplinary board.

| | |
|---|---|
| **Match words (EN)** | surgery planning, tumour board, multidisciplinary meeting, MDT, operation planning, consent, quadrantectomy, mastectomy, resection |
| **Match words (DE)** | Tumorboard, OP-Planung, Operationsplanung, Aufklärung, Quadrantektomie, Mastektomie, Resektion |
| **Implies** | none |

| Slot | Priority | Shape | Why it is here |
|---|---|---|---|
| Cancer type and histology [cancer_histology] | P1 | latest_value | Defines tumour type. |
| TNM stage and grade [tnm_grade] | P1 | latest_value | Stage and grade drive treatment. |
| Hormone receptors, HER2, Ki-67 [receptors_ki67] | P1 | latest_value | Biology of the tumour; surgeon and board need it. Hidden for the anaesthetist bundle. |
| Tumour side, quadrant and size [tumour_site_laterality] | P1 | laterality_finding | Side is the most expensive thing to get wrong; also feeds contradiction checks. |
| Tumour board recommendation [board_recommendation] | P1 | plan | The agreed treatment plan. |
| Planned procedure and date [planned_procedure] | P1 | plan | Anchors the task: procedure and date. |
| Breast and staging imaging [oncology_imaging] | P2 | list_all | Extent of disease. |
| BI-RADS category [birads] | P2 | latest_value | Imaging suspicion category; can disagree with histology wording. |
| Axillary and sentinel status [axillary_status] | P2 | latest_value | Nodal staging. |
| Earlier cancer treatment [prior_oncologic_tx] | P2 | list_all | Earlier treatment, including radiotherapy. |
| All procedures and operations [procedures_all] | P2 | list_all | Earlier operations and interventions. |
| Antiplatelets and anticoagulants [med_antithrombotic] | P2 | current_list | Bleeding and thrombosis management. |
| Allergies and intolerances, with reaction [allergies] | P2 | checklist | Safety-critical; never hidden. |
| All diagnoses with dates and status [diagnoses_all] | P2 | list_all | Comorbidity context. |
| Age and sex [age_sex] | P2 | latest_value | Age and sex frame risk, dosing and screening. |
| Diabetes: type and duration [diabetes_dx] | P3 | latest_value | Type and duration frame all diabetes care. |
| LVEF and echo findings [lvef] | P3 | latest_value | Cardiac pump function. |
| Weight [weight] | P3 | trend | Dosing, nutrition and disease trajectory. |
| Height and BMI [height_bmi] | P3 | latest_value | BMI for dosing, anaesthesia and metabolic risk. |

| | |
|---|---|
| **Expected but may be missing** | Axillary and sentinel status, Breast and staging imaging, Earlier cancer treatment |
| **Link rules** | R01, R17, R19 |
| **Guideline sources** | G11, G21, G17 |
| **Design note** | Receptors and Ki-67 are P1 here and absent from B01: this contrast is the demo of 'the screen changes with the role'. |

## B03 Diabetes review

Primary-care or specialist review of diabetes control, drugs and complications.

| | |
|---|---|
| **Match words (EN)** | diabetes review, diabetes check, glucose control, HbA1c review, diabetic, sugar, glycaemic |
| **Match words (DE)** | Diabeteskontrolle, Diabetes, Zuckerkrankheit, Blutzuckereinstellung, HbA1c-Kontrolle |
| **Implies** | none |

| Slot | Priority | Shape | Why it is here |
|---|---|---|---|
| Glucose-lowering drugs, incl. insulin [med_antidiabetic] | P1 | current_list | Glycaemic control and drug safety. |
| HbA1c: value, unit, date [hba1c] | P1 | latest_value | Latest value with its date; say plainly if nothing newer is documented. |
| Weight [weight] | P1 | trend | Dated weights and a calculated change, each input linked. |
| Diabetes: type and duration [diabetes_dx] | P1 | latest_value | Type and duration frame all diabetes care. |
| Blood glucose [glucose] | P2 | trend | Acute glycaemic state. |
| Creatinine and eGFR [kidney_function] | P2 | latest_value | Drug dosing and contrast safety. |
| Blood pressure and heart rate [vitals_bp_hr] | P2 | trend | Baseline haemodynamics. |
| Blood pressure drugs [med_antihypertensive] | P2 | current_list | Blood pressure control and interactions. |
| Lipid-lowering drugs [med_lipid] | P2 | current_list | Cardiovascular risk management. |
| Lipid profile [lipids] | P2 | latest_value | Vascular risk. |
| Diabetic complications [diabetic_complications] | P2 | list_all | End-organ involvement. |
| Hypoglycaemia [hypoglycaemia_hx] | P2 | event_history | Safety with glucose-lowering drugs. |
| Eye symptoms and eye follow-up [eye_history] | P2 | event_history | Visual events and eye follow-up. |
| Stopped, paused or changed medication [med_stopped_changed] | P2 | event_history | Changes over time; avoids acting on outdated lists. |
| All diagnoses with dates and status [diagnoses_all] | P2 | list_all | Comorbidity context. |
| Smoking and alcohol [smoking_alcohol] | P3 | latest_value | Perioperative and vascular risk modifiers. |
| Diet, appetite, swallowing [diet_nutrition] | P3 | latest_value | Nutritional status. |
| Mobility and functional capacity [mobility_function] | P3 | latest_value | Functional reserve and rehab needs. |
| Hospital stays, admission to discharge [hospital_stays] | P3 | list_all | Index events and timeline. |

| | |
|---|---|
| **Expected but may be missing** | Creatinine and eGFR, Lipid profile, Diabetic complications, Eye symptoms and eye follow-up, Blood glucose |
| **Link rules** | R14, R16, R21 |
| **Guideline sources** | G05, G10, G19 |
| **Design note** | Weight is P1 because unexplained change matters in diabetes review; the change is computed in code. |

## B04 Acute neurological or visual presentation (emergency)

Emergency assessment of dizziness, vision loss, weakness or speech disturbance.

| | |
|---|---|
| **Match words (EN)** | emergency, ED, A&E, night duty, dizziness, blurred vision, vision loss, TIA, stroke, weakness, speech, collapse |
| **Match words (DE)** | Notaufnahme, Nacht, Schwindel, Sehstörung, Visusverlust, Schlaganfall, TIA, Lähmung, Sprachstörung |
| **Implies** | B05 |

| Slot | Priority | Shape | Why it is here |
|---|---|---|---|
| Earlier similar episodes and their cause [prior_similar_episode] | P1 | event_history | Has this happened before, and what was concluded. |
| Stroke, TIA, amaurosis fugax [stroke_tia_hx] | P1 | event_history | Cerebrovascular events. |
| Carotid stenosis: side, degree, method, date [carotid_stenosis] | P1 | laterality_finding | Show side and degree next to the symptomatic side so the clinician can compare. |
| Antiplatelets and anticoagulants [med_antithrombotic] | P1 | current_list | Bleeding and thrombosis management. |
| Blood pressure drugs [med_antihypertensive] | P1 | current_list | Blood pressure control and interactions. |
| Allergies and intolerances, with reaction [allergies] | P1 | checklist | Safety-critical; never hidden. |
| Hypertension and hypertensive crises [hypertension_hx] | P2 | event_history | Blood pressure history and crises. |
| ECG and rhythm [ecg_rhythm] | P2 | latest_value | Rhythm and conduction baseline. |
| Diabetes: type and duration [diabetes_dx] | P2 | latest_value | Type and duration frame all diabetes care. |
| Blood glucose [glucose] | P2 | trend | Acute glycaemic state. |
| HbA1c: value, unit, date [hba1c] | P2 | latest_value | Long-term glycaemic control, always with date. |
| LVEF and echo findings [lvef] | P2 | latest_value | Cardiac pump function. |
| Vascular imaging: duplex, CT/MR angiography [vascular_imaging] | P2 | list_all | Imaging history, including whether contrast was used. |
| Head imaging, CT or MRI [head_imaging] | P2 | list_all | Intracranial imaging history. |
| Neurological findings and deficits [neuro_deficits] | P2 | latest_value | Neurological baseline. |
| Eye symptoms and eye follow-up [eye_history] | P2 | event_history | Visual events and eye follow-up. |
| Hospital stays, admission to discharge [hospital_stays] | P2 | list_all | Index events and timeline. |
| Blood pressure and heart rate [vitals_bp_hr] | P2 | trend | Baseline haemodynamics. |
| Advance directive / resuscitation status [code_status] | P2 | latest_value | Goals of care in emergencies. |
| Living situation and support [social_support] | P3 | latest_value | Discharge and rehab planning. |
| Smoking and alcohol [smoking_alcohol] | P3 | latest_value | Perioperative and vascular risk modifiers. |
| Current medication, all [med_current] | P3 | current_list | Complete current medication. |
| All diagnoses with dates and status [diagnoses_all] | P3 | list_all | Comorbidity context. |

| | |
|---|---|
| **Expected but may be missing** | Creatinine and eGFR, Blood glucose, ECG and rhythm, Neurological findings and deficits |
| **Link rules** | R02, R03, R04, R05, R09, R10, R11, R17, R21 |
| **Guideline sources** | G08, G17 |
| **Design note** | Implies B05: neurological symptoms usually lead to contrast imaging, so contrast safety appears without being asked. |

## B05 Safety check before iodinated contrast

Checklist before contrast-enhanced CT or angiography.

| | |
|---|---|
| **Match words (EN)** | contrast, iodinated contrast, CT angiography, CTA, contrast CT, contrast media, angiogram |
| **Match words (DE)** | Kontrastmittel, jodhaltig, CT-Angiographie, Angio-CT, KM-Gabe |
| **Implies** | none |

| Slot | Priority | Shape | Why it is here |
|---|---|---|---|
| Allergies and intolerances, with reaction [allergies] | P1 | checklist | Safety-critical; never hidden. |
| Vascular imaging: duplex, CT/MR angiography [vascular_imaging] | P1 | list_all | Earlier contrast studies, and whether contrast was avoided because of a reaction. |
| Glucose-lowering drugs, incl. insulin [med_antidiabetic] | P1 | current_list | Metformin and other agents that matter around contrast. |
| Creatinine and eGFR [kidney_function] | P1 | latest_value | Drug dosing and contrast safety. |
| Drugs relevant to kidney function [med_renal_relevant] | P2 | current_list | Drugs that affect or depend on kidney function. |
| Diabetes: type and duration [diabetes_dx] | P2 | latest_value | Type and duration frame all diabetes care. |
| Coronary disease, MI, stents, heart failure [coronary_hf] | P2 | list_all | Ischaemic and heart failure burden. |
| Age and sex [age_sex] | P2 | latest_value | Age and sex frame risk, dosing and screening. |
| Hypertension and hypertensive crises [hypertension_hx] | P3 | event_history | Blood pressure history and crises. |
| Weight [weight] | P3 | trend | Dosing, nutrition and disease trajectory. |
| Hospital stays, admission to discharge [hospital_stays] | P3 | list_all | Index events and timeline. |

| | |
|---|---|
| **Expected but may be missing** | Creatinine and eGFR |
| **Link rules** | R02, R03, R04, R05 |
| **Guideline sources** | G12, G13, G14 |
| **Design note** | A composite slot group: each item is found, documented as none, or not documented, independently. |

## B06 Open follow-ups after a hospital stay

Checking which recommendations from an earlier stay were done or are still open.

| | |
|---|---|
| **Match words (EN)** | follow-up, follow ups, open items, overdue, discharge recommendations, outstanding, still open |
| **Match words (DE)** | Nachkontrolle, Kontrolltermin, offen, Entlassungsempfehlung, ausstehend |
| **Implies** | none |

| Slot | Priority | Shape | Why it is here |
|---|---|---|---|
| Hospital stays, admission to discharge [hospital_stays] | P1 | list_all | Index events and timeline. |
| Follow-ups recommended at discharge [discharge_followups] | P1 | list_all | What was meant to happen next. |
| Evidence that a follow-up took place [followup_done] | P1 | status_check | No evidence found = 'not documented' (not 'no'). |
| Carotid stenosis: side, degree, method, date [carotid_stenosis] | P1 | laterality_finding | Side of the finding the follow-up was for. |
| Actions handed to the GP [gp_actions] | P2 | list_all | Tasks handed to primary care. |
| Vascular imaging: duplex, CT/MR angiography [vascular_imaging] | P2 | list_all | Imaging history, including whether contrast was used. |
| All diagnoses with dates and status [diagnoses_all] | P2 | list_all | Comorbidity context. |

| Slot | Priority | Shape | Why it is here |
|---|---|---|---|
| Eye symptoms and eye follow-up [eye_history] | P2 | event_history | Visual events and eye follow-up. |
| Stopped, paused or changed medication [med_stopped_changed] | P3 | event_history | Changes over time; avoids acting on outdated lists. |

| | |
|---|---|
| **Expected but may be missing** | Evidence that a follow-up took place |
| **Link rules** | R15, R21 |
| **Guideline sources** | G18 |
| **Design note** | Mostly a workflow task, so the support is local discharge policy, not a clinical guideline. |

## B07 Medication reconciliation and review

Building an accurate, complete medication list with doses, changes and safety context.

| | |
|---|---|
| **Match words (EN)** | medication review, reconciliation, drug chart, prescribing, polypharmacy, medicines check, med rec |
| **Match words (DE)** | Medikationsreview, Medikationsabgleich, Medikamentenliste, Polypharmazie |
| **Implies** | none |

| Slot | Priority | Shape | Why it is here |
|---|---|---|---|
| Current medication, all [med_current] | P1 | current_list | Complete current medication. |
| Stopped, paused or changed medication [med_stopped_changed] | P1 | event_history | Changes over time; avoids acting on outdated lists. |
| Allergies and intolerances, with reaction [allergies] | P1 | checklist | Safety-critical; never hidden. |
| Antiplatelets and anticoagulants [med_antithrombotic] | P2 | current_list | Bleeding and thrombosis management. |
| Glucose-lowering drugs, incl. insulin [med_antidiabetic] | P2 | current_list | Glycaemic control and drug safety. |
| Blood pressure drugs [med_antihypertensive] | P2 | current_list | Blood pressure control and interactions. |
| Lipid-lowering drugs [med_lipid] | P2 | current_list | Cardiovascular risk management. |
| Other cardiac drugs [med_cardiac_other] | P2 | current_list | Rhythm, rate and volume management. |
| Drugs relevant to kidney function [med_renal_relevant] | P2 | current_list | Drugs that affect or depend on kidney function. |
| Analgesics and psychotropic drugs [med_analgesic_psychotropic] | P2 | current_list | Sedation, interaction and bleeding relevance. |
| Creatinine and eGFR [kidney_function] | P2 | latest_value | Drug dosing and contrast safety. |
| Weight [weight] | P2 | trend | Dosing, nutrition and disease trajectory. |
| Age and sex [age_sex] | P2 | latest_value | Age and sex frame risk, dosing and screening. |
| All diagnoses with dates and status [diagnoses_all] | P2 | list_all | Comorbidity context. |
| Liver function [liver_function] | P3 | latest_value | Drug metabolism. |
| Hospital stays, admission to discharge [hospital_stays] | P3 | list_all | Index events and timeline. |

| | |
|---|---|
| **Expected but may be missing** | Creatinine and eGFR, Liver function |
| **Link rules** | R16, R17, R21 |
| **Guideline sources** | G15, G13 |
| **Design note** | Class slots (P2) are filtered views of med_current by the drug-class lookup, not separate searches. |

## B08 Admission or discharge handover

Ward-level summary for nursing admission, discharge planning or handover.

| | |
|---|---|
| **Match words (EN)** | admission, discharge, handover, ward, nursing assessment, going home, transfer |
| **Match words (DE)** | Aufnahme, Entlassung, Übergabe, Station, Pflegeanamnese, Transfer |
| **Implies** | none |

| Slot | Priority | Shape | Why it is here |
|---|---|---|---|
| All diagnoses with dates and status [diagnoses_all] | P1 | list_all | Comorbidity context. |
| Current medication, all [med_current] | P1 | current_list | Complete current medication. |
| Allergies and intolerances, with reaction [allergies] | P1 | checklist | Safety-critical; never hidden. |
| Follow-ups recommended at discharge [discharge_followups] | P1 | list_all | What was meant to happen next. |
| Mobility and functional capacity [mobility_function] | P1 | latest_value | Functional reserve and rehab needs. |
| Living situation and support [social_support] | P2 | latest_value | Discharge and rehab planning. |
| Advance directive / resuscitation status [code_status] | P2 | latest_value | Goals of care in emergencies. |
| Diet, appetite, swallowing [diet_nutrition] | P2 | latest_value | Nutritional status. |
| All procedures and operations [procedures_all] | P2 | list_all | Earlier operations and interventions. |
| Hospital stays, admission to discharge [hospital_stays] | P2 | list_all | Index events and timeline. |
| Weight [weight] | P3 | trend | Dosing, nutrition and disease trajectory. |

| | |
|---|---|
| **Expected but may be missing** | Mobility and functional capacity, Living situation and support, Advance directive / resuscitation status, Diet, appetite, swallowing |
| **Link rules** | R17, R21 |
| **Guideline sources** | G18 |

## B09 Hypertension and cardiovascular review

Blood pressure control and cardiovascular risk review.

| | |
|---|---|
| **Match words (EN)** | hypertension review, blood pressure review, cardiology review, heart check, cardiovascular risk |
| **Match words (DE)** | Hypertonie, Blutdruckkontrolle, Kardiologie, Herzkontrolle, kardiovaskuläres Risiko |
| **Implies** | none |

| Slot | Priority | Shape | Why it is here |
|---|---|---|---|
| Blood pressure drugs [med_antihypertensive] | P1 | current_list | Blood pressure control and interactions. |
| Blood pressure and heart rate [vitals_bp_hr] | P1 | trend | Baseline haemodynamics. |
| Hypertension and hypertensive crises [hypertension_hx] | P1 | event_history | Blood pressure history and crises. |
| LVEF and echo findings [lvef] | P2 | latest_value | Cardiac pump function. |
| Carotid stenosis: side, degree, method, date [carotid_stenosis] | P2 | laterality_finding | Side and degree guide imaging and symptom attribution. |
| Creatinine and eGFR [kidney_function] | P2 | latest_value | Drug dosing and contrast safety. |
| Lipid-lowering drugs [med_lipid] | P2 | current_list | Cardiovascular risk management. |
| Lipid profile [lipids] | P2 | latest_value | Vascular risk. |

| Slot | Priority | Shape | Why it is here |
|---|---|---|---|
| Stroke, TIA, amaurosis fugax [stroke_tia_hx] | P2 | event_history | Cerebrovascular events. |
| Diabetes: type and duration [diabetes_dx] | P2 | latest_value | Type and duration frame all diabetes care. |
| ECG and rhythm [ecg_rhythm] | P2 | latest_value | Rhythm and conduction baseline. |
| Coronary disease, MI, stents, heart failure [coronary_hf] | P2 | list_all | Ischaemic and heart failure burden. |
| Smoking and alcohol [smoking_alcohol] | P3 | latest_value | Perioperative and vascular risk modifiers. |
| Valve disease, prosthetic valves [valves] | P3 | latest_value | Structural heart disease. |

| | |
|---|---|
| **Expected but may be missing** | Creatinine and eGFR, Lipid profile, ECG and rhythm |
| **Link rules** | R21 |
| **Guideline sources** | G09, G07, G19 |

## B10 Weight change and nutrition review

Unexplained or documented weight change, with the context that could explain it.

| | |
|---|---|
| **Match words (EN)** | weight loss, weight change, unintentional weight loss, nutrition, appetite, BMI review, dietitian |
| **Match words (DE)** | Gewichtsverlust, Gewichtsabnahme, Ernährung, Appetit, Diätberatung |
| **Implies** | none |

| Slot | Priority | Shape | Why it is here |
|---|---|---|---|
| Weight [weight] | P1 | trend | Dosing, nutrition and disease trajectory. |
| Height and BMI [height_bmi] | P1 | latest_value | BMI for dosing, anaesthesia and metabolic risk. |
| All diagnoses with dates and status [diagnoses_all] | P1 | list_all | Comorbidity context. |
| Glucose-lowering drugs, incl. insulin [med_antidiabetic] | P2 | current_list | Glycaemic control and drug safety. |
| HbA1c: value, unit, date [hba1c] | P2 | latest_value | Long-term glycaemic control, always with date. |
| Blood glucose [glucose] | P2 | trend | Acute glycaemic state. |
| Diet, appetite, swallowing [diet_nutrition] | P2 | latest_value | Nutritional status. |
| Hospital stays, admission to discharge [hospital_stays] | P2 | list_all | Index events and timeline. |
| Earlier cancer treatment [prior_oncologic_tx] | P2 | list_all | Earlier treatment, including radiotherapy. |
| Liver function [liver_function] | P3 | latest_value | Drug metabolism. |
| Mobility and functional capacity [mobility_function] | P3 | latest_value | Functional reserve and rehab needs. |

| | |
|---|---|
| **Expected but may be missing** | Diet, appetite, swallowing, Height and BMI |
| **Link rules** | R14, R21 |
| **Guideline sources** | G18 |
| **Design note** | Shows context, never a cause: causes are for the clinician. |

## B11 Systemic cancer therapy planning

Oncologist deciding on adjuvant or systemic treatment.

| | |
|---|---|
| **Match words (EN)** | adjuvant, chemotherapy, endocrine therapy, systemic therapy, oncology review, hormone therapy |
| **Match words (DE)** | adjuvant, Chemotherapie, endokrine Therapie, Systemtherapie, Onkologie |
| **Implies** | none |

| Slot | Priority | Shape | Why it is here |
|---|---|---|---|
| Hormone receptors, HER2, Ki-67 [receptors_ki67] | P1 | latest_value | Drive systemic therapy choice. |
| Cancer type and histology [cancer_histology] | P1 | latest_value | Defines tumour type. |
| TNM stage and grade [tnm_grade] | P1 | latest_value | Stage and grade drive treatment. |
| Age and sex [age_sex] | P1 | latest_value | Age and sex frame risk, dosing and screening. |
| LVEF and echo findings [lvef] | P1 | latest_value | Cardiac pump function. |
| Planned procedure and date [planned_procedure] | P1 | plan | Anchors the task: procedure and date. |
| Creatinine and eGFR [kidney_function] | P2 | latest_value | Drug dosing and contrast safety. |
| Liver function [liver_function] | P2 | latest_value | Drug metabolism. |
| Haemoglobin and platelets [blood_count] | P2 | latest_value | Anaemia and platelet reserve. |
| Earlier cancer treatment [prior_oncologic_tx] | P2 | list_all | Earlier treatment, including radiotherapy. |
| Hypertension and hypertensive crises [hypertension_hx] | P2 | event_history | Blood pressure history and crises. |
| Diabetes: type and duration [diabetes_dx] | P2 | latest_value | Type and duration frame all diabetes care. |
| Current medication, all [med_current] | P2 | current_list | Complete current medication. |
| Allergies and intolerances, with reaction [allergies] | P2 | checklist | Safety-critical; never hidden. |
| Weight [weight] | P2 | trend | Dosing, nutrition and disease trajectory. |
| Height and BMI [height_bmi] | P2 | latest_value | BMI for dosing, anaesthesia and metabolic risk. |
| Mobility and functional capacity [mobility_function] | P2 | latest_value | Functional reserve and rehab needs. |
| Axillary and sentinel status [axillary_status] | P3 | latest_value | Nodal staging. |
| Breast and staging imaging [oncology_imaging] | P3 | list_all | Extent of disease. |
| BI-RADS category [birads] | P3 | latest_value | Imaging suspicion category; can disagree with histology wording. |

| | |
|---|---|
| **Expected but may be missing** | Creatinine and eGFR, Liver function, Haemoglobin and platelets, Mobility and functional capacity |
| **Link rules** | R12, R17, R21 |
| **Guideline sources** | G11, G21, G17 |

## B12 Radiotherapy planning (incl. intra-operative)

Radiation oncologist planning treatment such as IORT.

| | |
|---|---|
| **Match words (EN)** | radiotherapy, IORT, intraoperative radiotherapy, radiation, irradiation, boost |
| **Match words (DE)** | Strahlentherapie, IORT, intraoperative Radiotherapie, Bestrahlung |
| **Implies** | none |

| Slot | Priority | Shape | Why it is here |
|---|---|---|---|
| Tumour side, quadrant and size [tumour_site_laterality] | P1 | laterality_finding | Side and site; critical for surgery and radiotherapy. |
| Planned procedure and date [planned_procedure] | P1 | plan | Anchors the task: procedure and date. |
| Tumour board recommendation [board_recommendation] | P1 | plan | The agreed treatment plan. |
| TNM stage and grade [tnm_grade] | P1 | latest_value | Stage and grade drive treatment. |
| Earlier cancer treatment [prior_oncologic_tx] | P1 | list_all | Earlier treatment, including radiotherapy. |
| Pacemaker or ICD [cardiac_device] | P1 | latest_value | Devices affect procedures and radiotherapy. |
| Cancer type and histology [cancer_histology] | P2 | latest_value | Defines tumour type. |
| Breast and staging imaging [oncology_imaging] | P2 | list_all | Extent of disease. |
| BI-RADS category [birads] | P2 | latest_value | Imaging suspicion category; can disagree with histology wording. |
| Axillary and sentinel status [axillary_status] | P2 | latest_value | Nodal staging. |
| Coronary disease, MI, stents, heart failure [coronary_hf] | P2 | list_all | Ischaemic and heart failure burden. |
| All procedures and operations [procedures_all] | P2 | list_all | Earlier operations and interventions. |
| LVEF and echo findings [lvef] | P3 | latest_value | Cardiac pump function. |
| Allergies and intolerances, with reaction [allergies] | P3 | checklist | Safety-critical; never hidden. |
| Age and sex [age_sex] | P3 | latest_value | Age and sex frame risk, dosing and screening. |

| | |
|---|---|
| **Expected but may be missing** | Pacemaker or ICD, Earlier cancer treatment |
| **Link rules** | R13, R21 |
| **Guideline sources** | G11, G21, G17 |

## B13 Post-operative rehabilitation

Physiotherapy or rehab after surgery.

| | |
|---|---|
| **Match words (EN)** | rehabilitation, physiotherapy, post-operative, mobilisation, rehab, shoulder, arm |
| **Match words (DE)** | Rehabilitation, Physiotherapie, postoperativ, Mobilisation |
| **Implies** | none |

| Slot | Priority | Shape | Why it is here |
|---|---|---|---|
| Planned procedure and date [planned_procedure] | P1 | plan | Anchors the task: procedure and date. |
| Tumour side, quadrant and size [tumour_site_laterality] | P1 | laterality_finding | Side and site; critical for surgery and radiotherapy. |
| Mobility and functional capacity [mobility_function] | P1 | latest_value | Functional reserve and rehab needs. |
| LVEF and echo findings [lvef] | P2 | latest_value | Cardiac pump function. |
| Blood pressure and heart rate [vitals_bp_hr] | P2 | trend | Baseline haemodynamics. |
| Antiplatelets and anticoagulants [med_antithrombotic] | P2 | current_list | Bleeding and thrombosis management. |
| All diagnoses with dates and status [diagnoses_all] | P2 | list_all | Comorbidity context. |
| Stroke, TIA, amaurosis fugax [stroke_tia_hx] | P2 | event_history | Cerebrovascular events. |
| Living situation and support [social_support] | P2 | latest_value | Discharge and rehab planning. |

| Slot | Priority | Shape | Why it is here |
|---|---|---|---|
| Diabetes: type and duration [diabetes_dx] | P3 | latest_value | Type and duration frame all diabetes care. |
| Carotid stenosis: side, degree, method, date [carotid_stenosis] | P3 | laterality_finding | Side and degree guide imaging and symptom attribution. |

| | |
|---|---|
| **Expected but may be missing** | Mobility and functional capacity, Living situation and support |
| **Link rules** | R21 |
| **Guideline sources** | G18 |

## B14 Eye or visual-event review

Ophthalmology review of visual symptoms and eye follow-up.

| | |
|---|---|
| **Match words (EN)** | ophthalmology, eye review, vision loss, transient vision loss, amaurosis, retina, eye check |
| **Match words (DE)** | Augenarzt, Ophthalmologie, Visusverlust, Amaurosis, Netzhaut, Augenkontrolle |
| **Implies** | none |

| Slot | Priority | Shape | Why it is here |
|---|---|---|---|
| Eye symptoms and eye follow-up [eye_history] | P1 | event_history | Visual events and eye follow-up. |
| Stroke, TIA, amaurosis fugax [stroke_tia_hx] | P1 | event_history | Cerebrovascular events. |
| Carotid stenosis: side, degree, method, date [carotid_stenosis] | P1 | laterality_finding | Side and degree guide imaging and symptom attribution. |
| Diabetes: type and duration [diabetes_dx] | P2 | latest_value | Type and duration frame all diabetes care. |
| HbA1c: value, unit, date [hba1c] | P2 | latest_value | Long-term glycaemic control, always with date. |
| Diabetic complications [diabetic_complications] | P2 | list_all | End-organ involvement. |
| Hypertension and hypertensive crises [hypertension_hx] | P2 | event_history | Blood pressure history and crises. |
| Blood pressure and heart rate [vitals_bp_hr] | P2 | trend | Baseline haemodynamics. |
| Antiplatelets and anticoagulants [med_antithrombotic] | P2 | current_list | Bleeding and thrombosis management. |
| Follow-ups recommended at discharge [discharge_followups] | P2 | list_all | What was meant to happen next. |
| Lipid profile [lipids] | P3 | latest_value | Vascular risk. |

| | |
|---|---|
| **Expected but may be missing** | Diabetic complications, Eye symptoms and eye follow-up |
| **Link rules** | R09, R21 |
| **Guideline sources** | G05, G08 |

## B15 Minor procedure and bleeding check (e.g. dental)

Dentist or minor-procedure clinician checking bleeding and infection-relevant history.

| | |
|---|---|
| **Match words (EN)** | dental, tooth extraction, minor procedure, local anaesthetic, dentist, biopsy |
| **Match words (DE)** | Zahnarzt, Zahnextraktion, kleiner Eingriff, Lokalanästhesie |
| **Implies** | none |

| Slot | Priority | Shape | Why it is here |
|---|---|---|---|
| Antiplatelets and anticoagulants [med_antithrombotic] | P1 | current_list | Bleeding and thrombosis management. |

| Slot | Priority | Shape | Why it is here |
|---|---|---|---|
| Allergies and intolerances, with reaction [allergies] | P1 | checklist | Safety-critical; never hidden. |
| Valve disease, prosthetic valves [valves] | P1 | latest_value | Structural heart disease. |
| Diabetes: type and duration [diabetes_dx] | P2 | latest_value | Type and duration frame all diabetes care. |
| HbA1c: value, unit, date [hba1c] | P2 | latest_value | Long-term glycaemic control, always with date. |
| Haemoglobin and platelets [blood_count] | P2 | latest_value | Anaemia and platelet reserve. |
| Coagulation tests and bleeding history [coagulation] | P2 | latest_value | Bleeding risk. |
| Coronary disease, MI, stents, heart failure [coronary_hf] | P2 | list_all | Ischaemic and heart failure burden. |
| Current medication, all [med_current] | P2 | current_list | Complete current medication. |
| Analgesics and psychotropic drugs [med_analgesic_psychotropic] | P3 | current_list | Sedation, interaction and bleeding relevance. |

| | |
|---|---|
| **Expected but may be missing** | Coagulation tests and bleeding history, Valve disease, prosthetic valves |
| **Link rules** | R01, R17, R20 |
| **Guideline sources** | G03, G16, G18 |

## B16 Emergency or paramedic handover

Fast safety summary for pre-hospital or triage handover.

| | |
|---|---|
| **Match words (EN)** | paramedic, ambulance, triage, handover, collapse, found at home, pre-hospital |
| **Match words (DE)** | Rettungsdienst, Notarzt, Triage, Übergabe, Kollaps, präklinisch |
| **Implies** | none |

| Slot | Priority | Shape | Why it is here |
|---|---|---|---|
| Allergies and intolerances, with reaction [allergies] | P1 | checklist | Safety-critical; never hidden. |
| Antiplatelets and anticoagulants [med_antithrombotic] | P1 | current_list | Bleeding and thrombosis management. |
| All diagnoses with dates and status [diagnoses_all] | P1 | list_all | Comorbidity context. |
| Current medication, all [med_current] | P1 | current_list | Complete current medication. |
| Advance directive / resuscitation status [code_status] | P1 | latest_value | Goals of care in emergencies. |
| Age and sex [age_sex] | P1 | latest_value | Age and sex frame risk, dosing and screening. |
| Weight [weight] | P2 | trend | Dosing, nutrition and disease trajectory. |
| Mobility and functional capacity [mobility_function] | P2 | latest_value | Functional reserve and rehab needs. |
| Living situation and support [social_support] | P2 | latest_value | Discharge and rehab planning. |
| Earlier similar episodes and their cause [prior_similar_episode] | P2 | event_history | Recurrence and earlier work-up. |
| Hospital stays, admission to discharge [hospital_stays] | P3 | list_all | Index events and timeline. |

| | |
|---|---|
| **Expected but may be missing** | Advance directive / resuscitation status, Living situation and support |
| **Link rules** | R17, R21 |
| **Guideline sources** | G18 |

## B17 Carotid and vascular review

Vascular or neurovascular review of carotid disease.

| | |
|---|---|
| **Match words (EN)** | carotid, endarterectomy, carotid stenosis, vascular surgery, vascular review, duplex |
| **Match words (DE)** | Karotis, Endarteriektomie, Karotisstenose, Gefäßchirurgie, Duplex |
| **Implies** | none |

| Slot | Priority | Shape | Why it is here |
|---|---|---|---|
| Carotid stenosis: side, degree, method, date [carotid_stenosis] | P1 | laterality_finding | Side and degree guide imaging and symptom attribution. |
| Stroke, TIA, amaurosis fugax [stroke_tia_hx] | P1 | event_history | Cerebrovascular events. |
| Vascular imaging: duplex, CT/MR angiography [vascular_imaging] | P1 | list_all | Imaging history, including whether contrast was used. |
| Antiplatelets and anticoagulants [med_antithrombotic] | P1 | current_list | Bleeding and thrombosis management. |
| Lipid-lowering drugs [med_lipid] | P1 | current_list | Cardiovascular risk management. |
| Blood pressure and heart rate [vitals_bp_hr] | P2 | trend | Baseline haemodynamics. |
| Hypertension and hypertensive crises [hypertension_hx] | P2 | event_history | Blood pressure history and crises. |
| Diabetes: type and duration [diabetes_dx] | P2 | latest_value | Type and duration frame all diabetes care. |
| Smoking and alcohol [smoking_alcohol] | P2 | latest_value | Perioperative and vascular risk modifiers. |
| Follow-ups recommended at discharge [discharge_followups] | P2 | list_all | What was meant to happen next. |
| Evidence that a follow-up took place [followup_done] | P2 | status_check | Whether it did happen. Absence is shown as 'not documented'. |
| Neurological findings and deficits [neuro_deficits] | P2 | latest_value | Neurological baseline. |
| Eye symptoms and eye follow-up [eye_history] | P2 | event_history | Visual events and eye follow-up. |
| LVEF and echo findings [lvef] | P3 | latest_value | Cardiac pump function. |

| | |
|---|---|
| **Expected but may be missing** | Evidence that a follow-up took place, Smoking and alcohol, Neurological findings and deficits |
| **Link rules** | R09, R10, R19, R21 |
| **Guideline sources** | G08, G17 |

## B18 General core summary (fallback)

Shown when the role or task is not recognised. A visible note says so.

| | |
|---|---|
| **Match words (EN)** | (fallback; no tags) |
| **Match words (DE)** | (Fallback; keine Tags) |
| **Implies** | none |

| Slot | Priority | Shape | Why it is here |
|---|---|---|---|
| Age and sex [age_sex] | P1 | latest_value | Age and sex frame risk, dosing and screening. |
| All diagnoses with dates and status [diagnoses_all] | P1 | list_all | Comorbidity context. |
| Current medication, all [med_current] | P1 | current_list | Complete current medication. |
| Allergies and intolerances, with reaction [allergies] | P1 | checklist | Safety-critical; never hidden. |
| All procedures and operations [procedures_all] | P2 | list_all | Earlier operations and interventions. |

| Slot | Priority | Shape | Why it is here |
|---|---|---|---|
| Hospital stays, admission to discharge [hospital_stays] | P2 | list_all | Index events and timeline. |
| Weight [weight] | P2 | trend | Dosing, nutrition and disease trajectory. |
| Blood pressure and heart rate [vitals_bp_hr] | P2 | trend | Baseline haemodynamics. |
| Living situation and support [social_support] | P3 | latest_value | Discharge and rehab planning. |
| Advance directive / resuscitation status [code_status] | P3 | latest_value | Goals of care in emergencies. |

| | |
|---|---|
| **Expected but may be missing** | n/a |
| **Link rules** | R17, R21 |
| **Guideline sources** | G18 |
| **Design note** | Never a silent fallback: the screen states 'Role/task not recognised, showing general summary'. |

# 5 Fact slot catalogue with German search hints

Hints feed the offline extractor prompt and a lexical pre-filter over the German documents. They are aids, not rules: extraction must still read the passage and the validator must still find the quote.

| Slot ID | Label | Shape | German search hints | Codes (verify) | Data notes |
|---|---|---|---|---|---|
| **BASICS** | | | | | |
| age_sex | Age and sex | latest_value | Geburtsdatum, geb., Alter, weiblich, männlich, Patientin, Patient | | Compute age from the date of birth at the anchor date (57 for the demo patient: born 15 March 1959, anchor 15 April 2016). patient.json stores today's age (67): never use it. |
| weight | Weight | trend | Gewicht, Körpergewicht, kg, Gewichtsabnahme, Gewichtsverlust | | Extraction dates are month precision; order values inside a month by document date. |
| height_bmi | Height and BMI | latest_value | Größe, Körpergröße, cm, BMI, Body-Mass-Index | | |
| vitals_bp_hr | Blood pressure and heart rate | trend | RR, Blutdruck, mmHg, Puls, Herzfrequenz, HF | | |
| spo2_resp | Oxygen saturation, respiratory rate, temperature | latest_value | SpO2, Sättigung, Atemfrequenz, Temperatur, Fieber | | |
| allergies | Allergies and intolerances, with reaction | checklist | Allergie, Allergien, Unverträglichkeit, Kontrastmittel, Penicillin, Latex, keine bekannten Allergien | | cave.json holds one vague entry ('Arzneimittel-Allergie') with no documentIds: use it only as a search trigger and extract the allergen and reaction from the letter text, otherwise the link fails traceability. |
| code_status | Advance directive / resuscitation status | latest_value | Patientenverfügung, Reanimation, DNR, Vorsorgevollmacht | | |
| social_support | Living situation and support | latest_value | wohnt, allein, Angehörige, Pflege, Heim, Selbständigkeit | | |
| mobility_function | Mobility and functional capacity | latest_value | Mobilität, Gehhilfe, Belastbarkeit, Treppensteigen, selbständig, Sturz | | |
| smoking_alcohol | Smoking and alcohol | latest_value | Nikotin, Raucherin, Nichtraucher, pack years, Alkohol | | |
| **HISTORY** | | | | | |
| diagnoses_all | All diagnoses with dates and status | list_all | Diagnosen, Entlassungsdiagnosen, Nebendiagnosen, Anamnese, Z.n., Zustand nach | | IBM statuses: positive, negative (ruled out), suspected, statuspost. Show negative as 'ruled out', suspected as 'suspected', statuspost as 'past'; never list them as confirmed facts. |
| procedures_all | All procedures and operations | list_all | Operation, OP, Eingriff, Intervention, Biopsie, Z.n. | | IBM statuses: positive (done), statuspost, recommended. Only 9 of 62 procedure dates have a day. 'recommended' feeds board and follow-up slots; never show it as done. |
| hospital_stays | Hospital stays, admission to discharge | list_all | Aufenthalt, stationär, aufgenommen, entlassen, Entlassung, Aufnahme | | documents.json stay dates can differ from the letter text; the text wins. |
| prior_similar_episode | Earlier similar episodes and their cause | event_history | Z.n., bereits, wiederholt, Ereignis, Voraufenthalt, Ursache, abgeklärt | | |
| discharge_followups | Follow-ups recommended at discharge | list_all | Empfehlung, Kontrolle, Wiedervorstellung, in 2 Monaten, Verlaufskontrolle, Procedere | | Candidates come from procedures marked 'recommended'; confirm the wording in the letter text. |

| Slot ID | Label | Shape | German search hints | Codes (verify) | Data notes |
|---|---|---|---|---|---|
| followup_done | Evidence that a follow-up took place | status_check | Kontrolle durchgeführt, Duplex, Befund vom, Wiedervorstellung | | |
| gp_actions | Actions handed to the GP | list_all | Hausarzt, niedergelassen, Weiterbetreuung, Anpassung durch Hausarzt | | |
| **MEDICATION** | | | | | |
| med_current | Current medication, all | current_list | Medikation, Dauermedikation, Entlassungsmedikation, Therapie, 1-0-1, mg, Dosierung | | Schedules have four numbers (morning, midday, evening, night); show three and decode, for example 1 0 0 0 -> 1-0-0 'once in the morning'. Dose sits inside IBM's matched text; re-read it from the canonical text. |
| med_antithrombotic | Antiplatelets and anticoagulants | current_list | ASS, Acetylsalicylsäure, Thrombo ASS, Aspirin, Clopidogrel, Plavix, Marcumar, Phenprocoumon, Xarelto, Eliquis, Pradaxa, Heparin, Enoxaparin, Antikoagulation, Thrombozytenaggregationshemmer | ATC B01 | |
| med_antidiabetic | Glucose-lowering drugs, incl. insulin | current_list | Metformin, Glucophage, Pioglitazon, Actos, Exenatid, Byetta, Insulin, Sitagliptin, Glimepirid, Antidiabetika | ATC A10 | |
| med_antihypertensive | Blood pressure drugs | current_list | Candesartan, Blopress, Ramipril, Amlodipin, Bisoprolol, Metoprolol, Hydrochlorothiazid, Antihypertensiva | ATC C02, C03, C07, C08, C09 | |
| med_lipid | Lipid-lowering drugs | current_list | Statin, Atorvastatin, Simvastatin, Ezetimib, Cholesterinsenker | ATC C10 | |
| med_cardiac_other | Other cardiac drugs | current_list | Betablocker, Amiodaron, Digoxin, Nitrat, Diuretikum, Furosemid, Antiarrhythmika | ATC C01 | |
| med_renal_relevant | Drugs relevant to kidney function | current_list | NSAR, Ibuprofen, Diclofenac, ACE-Hemmer, Sartane, Diuretika, Metformin | ATC M01A, C09, C03, A10BA | |
| med_analgesic_psychotropic | Analgesics and psychotropic drugs | current_list | Opioid, Tramadol, Paracetamol, Benzodiazepin, Antidepressiva, Neuroleptika | ATC N02, N05, N06 | |
| med_stopped_changed | Stopped, paused or changed medication | event_history | abgesetzt, pausiert, umgestellt, neu, reduziert, erhöht, Dosisanpassung | | |
| **CARDIOVASCULAR** | | | | | |
| lvef | LVEF and echo findings | latest_value | EF, LVEF, linksventrikuläre Funktion, Ejektionsfraktion, Echokardiographie, Herzecho, systolische Funktion | | |
| valves | Valve disease, prosthetic valves | latest_value | Klappe, Aortenstenose, Mitralinsuffizienz, Klappenersatz | ICD-10 I05-I08, I34-I39 | |
| ecg_rhythm | ECG and rhythm | latest_value | EKG, Sinusrhythmus, Vorhofflimmern, Schenkelblock, QT | ICD-10 I44-I49 | |
| coronary_hf | Coronary disease, MI, stents, heart failure | list_all | KHK, Infarkt, Stent, PCI, Bypass, Herzinsuffizienz, Angina | ICD-10 I20-I25, I50 | |
| cardiac_device | Pacemaker or ICD | latest_value | Schrittmacher, Herzschrittmacher, ICD, Defibrillator | | |

| Slot ID | Label | Shape | German search hints | Codes (verify) | Data notes |
|---|---|---|---|---|---|
| hypertension_hx | Hypertension and hypertensive crises | event_history | Hypertonie, hypertensive Krise, Blutdruckentgleisung, RR | ICD-10 I10-I15 | |
| **NEUROVASCULAR** | | | | | |
| carotid_stenosis | Carotid stenosis: side, degree, method, date | laterality_finding | Karotisstenose, A. carotis interna, ACI, Stenose, Duplex, Prozent, links, rechts, Stenosegrad | ICD-10 I65.2 | Laterality comes from the text only; IBM's matched fragments may be cut or carry markup. |
| stroke_tia_hx | Stroke, TIA, amaurosis fugax | event_history | Insult, Schlaganfall, TIA, Amaurosis fugax, Visusverlust, Sehstörung, Hemiparese, Aphasie | ICD-10 I60-I69, G45, H34.0 | |
| vascular_imaging | Vascular imaging: duplex, CT/MR angiography | list_all | Duplexsonographie, CT, CTA, MRA, Angiographie, mit Kontrastmittel, ohne Kontrastmittel | | |
| neuro_deficits | Neurological findings and deficits | latest_value | Parese, Sprache, Schwindel, Gesichtsfeld, neurologischer Status | | |
| head_imaging | Head imaging, CT or MRI | list_all | Schädel-CT, cCT, MRT, Ischämie, Blutung | | |
| eye_history | Eye symptoms and eye follow-up | event_history | Auge, Sehverlust, Visus, Augenarzt, Fundus, Netzhaut | ICD-10 H00-H59; ATC S01 | |
| **METABOLIC AND LABS** | | | | | |
| diabetes_dx | Diabetes: type and duration | latest_value | Diabetes mellitus Typ 2, DM2, Diabetes seit | ICD-10 E10-E14 (E11 type 2) | |
| hba1c | HbA1c: value, unit, date | latest_value | HbA1c, glykiertes Hämoglobin, %, mmol/mol | | The only lab value documented in the record; show month and year. |
| glucose | Blood glucose | trend | Blutzucker, Glukose, BZ, nüchtern, mg/dl | | Empty lab table in both discharge letters: show 'not documented' with that note. |
| hypoglycaemia_hx | Hypoglycaemia | event_history | Hypoglykämie, Unterzuckerung | ICD-10 E16 | |
| diabetic_complications | Diabetic complications | list_all | Retinopathie, Nephropathie, Neuropathie, diabetischer Fuß | ICD-10 E10-E14 with .2-.5 | |
| lipids | Lipid profile | latest_value | Cholesterin, LDL, HDL, Triglyceride | ICD-10 E78 | Empty lab table in both discharge letters: show 'not documented' with that note. |
| kidney_function | Creatinine and eGFR | latest_value | Kreatinin, eGFR, GFR, Nierenfunktion, Niereninsuffizienz, Harnstoff | ICD-10 N17-N19 | Both discharge letters end with an empty lab table and no lab value except HbA1c is documented: show 'not documented' and link the empty table as the searched passage. |
| electrolytes | Electrolytes | latest_value | Kalium, Natrium, Elektrolyte | | Empty lab table in both discharge letters: show 'not documented' with that note. |
| blood_count | Haemoglobin and platelets | latest_value | Hämoglobin, Hb, Thrombozyten, Blutbild, Leukozyten | | Empty lab table in both discharge letters: show 'not documented' with that note. |
| coagulation | Coagulation tests and bleeding history | latest_value | INR, Quick, aPTT, Gerinnung, Blutungsneigung, Blutungsanamnese | | Empty lab table in both discharge letters: show 'not documented' with that note. |
| liver_function | Liver function | latest_value | GOT, GPT, AST, ALT, Bilirubin, GGT | ICD-10 K70-K77 | Empty lab table in both discharge letters: show 'not documented' with that note. |
| **ONCOLOGY** | | | | | |

| Slot ID | Label | Shape | German search hints | Codes (verify) | Data notes |
|---|---|---|---|---|---|
| cancer_histology | Cancer type and histology | latest_value | Karzinom, Mammakarzinom, invasiv duktal, Histologie, Stanzbiopsie | ICD-10 C50 | |
| tnm_grade | TNM stage and grade | latest_value | TNM, pT, cT, N0, M0, V0, G2, Grading, Stadium | ICD-10 C50 | |
| receptors_ki67 | Hormone receptors, HER2, Ki-67 | latest_value | Östrogenrezeptor, ER, Progesteronrezeptor, PR, HER2, Allred, IRS, Ki-67, Proliferationsrate | | |
| tumour_site_laterality | Tumour side, quadrant and size | laterality_finding | links, rechts, Quadrant, Lokalisation, mm, cm, Tumorgröße | | |
| oncology_imaging | Breast and staging imaging | list_all | Mammographie, Sonographie, MRT, Staging, Szintigraphie | | |
| birads | BI-RADS category | latest_value | BI-RADS, BIRADS, Mammographie, Mamma-Sonographie, Kategorie | | Present in scores-all.json (the data README says 5 means very likely cancer). Compare with histology wording; a mismatch is a contradiction candidate. |
| axillary_status | Axillary and sentinel status | latest_value | Sentinel, Wächterlymphknoten, Axilla, Lymphknoten | | |
| board_recommendation | Tumour board recommendation | plan | Tumorboard, Empfehlung, Quadrantektomie, IORT, Mastektomie | | Candidates come from procedures marked 'recommended'; confirm the wording in the letter text. |
| planned_procedure | Planned procedure and date | plan | geplant, Operationstermin, OP am, Termin | | |
| prior_oncologic_tx | Earlier cancer treatment | list_all | Chemotherapie, Bestrahlung, Radiotherapie, Hormontherapie, Z.n. | | |
| **PERIOPERATIVE** | | | | | |
| prior_anaesthesia | Earlier anaesthesia and complications | event_history | Narkose, Anästhesie, Komplikation, maligne Hyperthermie, PONV, schwieriger Atemweg | | |
| airway | Airway assessment | latest_value | Mallampati, Mundöffnung, Zahnstatus, Atemweg, Intubation | | |
| pulmonary_hx | Lung disease and sleep apnoea | list_all | COPD, Asthma, Dyspnoe, Lungenfunktion, Schlafapnoe, Thoraxröntgen | ICD-10 J40-J47, G47.3 | |
| diet_nutrition | Diet, appetite, swallowing | latest_value | Ernährung, Diät, Appetit, Schlucken, Dysphagie | | |

# 6 Drug class lookup

The extractor reads all medication once; this table tags each drug with a class and feeds the class slots. Generic names are primary. Brand names are examples to improve lexical matching and must be checked against Austrian drug references.

| Class | Generic names | Brand examples | Feeds slots |
|---|---|---|---|
| antiplatelet | acetylsalicylic acid (ASS), clopidogrel, prasugrel, ticagrelor, dipyridamole | Thrombo ASS, Aspirin, Plavix, Efient, Brilique, Persantin | med_antithrombotic |
| anticoagulant_vka | phenprocoumon, warfarin | Marcumar, Coumadin | med_antithrombotic |
| anticoagulant_doac | apixaban, rivaroxaban, dabigatran, edoxaban | Eliquis, Xarelto, Pradaxa, Lixiana | med_antithrombotic |
| heparin | enoxaparin, dalteparin, nadroparin, unfractionated heparin | Lovenox, Clexane, Fragmin, Fraxiparin | med_antithrombotic |
| biguanide | metformin | Glucophage | med_antidiabetic; med_renal_relevant |
| sulfonylurea | glimepiride, gliclazide, glibenclamide | Amaryl, Diamicron, Euglucon | med_antidiabetic |
| glitazone | pioglitazone | Actos | med_antidiabetic |
| dpp4_inhibitor | sitagliptin, vildagliptin, linagliptin, saxagliptin | Januvia, Galvus, Trajenta, Onglyza | med_antidiabetic |
| glp1_agonist | exenatide, liraglutide, dulaglutide, semaglutide | Byetta, Bydureon, Victoza, Trulicity, Ozempic | med_antidiabetic |
| sglt2_inhibitor | empagliflozin, dapagliflozin, canagliflozin | Jardiance, Forxiga, Invokana | med_antidiabetic |
| insulin | all insulins and analogues | (many brands) | med_antidiabetic |
| ace_inhibitor | ramipril, enalapril, perindopril | Delix, Tritace, Renitec, Prestarium | med_antihypertensive; med_renal_relevant |
| arb | candesartan, valsartan, losartan, telmisartan, irbesartan | Blopress, Atacand, Diovan, Cosaar, Micardis, Aprovel | med_antihypertensive; med_renal_relevant |
| beta_blocker | bisoprolol, metoprolol, nebivolol, carvedilol | Concor, Beloc, Nebilet, Dilatrend | med_antihypertensive; med_cardiac_other |
| calcium_channel_blocker | amlodipine, nifedipine, lercanidipine | Norvasc, Adalat, Zanidip | med_antihypertensive |
| diuretic | hydrochlorothiazide, furosemide, torasemide, spironolactone | Lasix, Unat, Aldactone | med_antihypertensive; med_cardiac_other; med_renal_relevant |
| statin | atorvastatin, simvastatin, rosuvastatin, pravastatin | Sortis, Zocor, Crestor | med_lipid |
| nsaid | ibuprofen, diclofenac, naproxen | Brufen, Voltaren | med_renal_relevant; med_analgesic_psychotropic |
| opioid | tramadol, morphine, oxycodone | (many brands) | med_analgesic_psychotropic |
| renin_inhibitor | aliskiren | (brands to check) | med_antihypertensive; med_renal_relevant |
| methotrexate | methotrexate | (brands to check) | med_current |
| benzodiazepine | lorazepam, alprazolam, diazepam | Temesta, Xanor, Valium | med_analgesic_psychotropic |

# 7 Link rules

Each rule says: when the task matches and a fact of this kind is found, also surface these slots. Rules show facts and the reason; they never print a clinical conclusion. All rules need clinician sign-off.

| ID | When the task is | And this fact is found | Surface | How to display | Sources |
|---|---|---|---|---|---|
| R01 | Procedure, surgery or anaesthesia | Antiplatelet or anticoagulant found | med_antithrombotic; coagulation; blood_count | Bleeding risk and technique choice. Show drug, dose, schedule, as-of date. Last dose is usually not documented. | G02,G03,G07 |
| R02 | Contrast imaging | Contrast reaction found in allergies | allergies; vascular_imaging | Show allergen and reaction, and whether earlier imaging was done without contrast for that reason. | G13,G14 |
| R03 | Contrast imaging | Metformin (biguanide) found | med_antidiabetic; kidney_function | Show dose and the latest kidney function, or 'not documented'. NICE NG148 (2024) says do not delay contrast in an emergency, so show the gap but never block the screen on it. | G12,G14 |
| R04 | Contrast imaging | Diabetes, heart failure or hypertension found | kidney_function; diabetes_dx; coronary_hf | Kidney function is the common thread; show it with date. | G12,G14 |
| R05 | Contrast imaging | ACE inhibitor, ARB, diuretic or NSAID found | med_renal_relevant; kidney_function | Show the drugs and the latest kidney function. | G12 |
| R06 | Anaesthesia or sedation | GLP-1 agonist found | med_antidiabetic | Show drug, dose and schedule. Fasting guidance for this class has changed in recent years, so cite the edition in force. | G18 (to source) |
| R07 | Surgery | Diabetes found | hba1c; med_antidiabetic; glucose | Show latest HbA1c with its date; NICE NG45 asks if it was tested within 3 months. | G01,G05 |
| R08 | Surgery or contrast | ACE inhibitor or ARB found | med_antihypertensive; vitals_bp_hr; kidney_function | Show the drug, blood pressure values and kidney function. | G02 |
| R09 | Neurological or visual symptoms | Carotid stenosis found | carotid_stenosis; stroke_tia_hx; vascular_imaging; eye_history | Show side and degree beside the symptomatic side. Compare sides in code and place both passages together. | G08 |
| R10 | Neurological symptoms | Antithrombotic found | med_antithrombotic | Show last documented antithrombotics with dose and as-of date. | G08 |
| R11 | Neurological symptoms | Any | hypertension_hx; diabetes_dx; glucose; ecg_rhythm; stroke_tia_hx | Show history of crises, glucose-affecting disease, rhythm and earlier events. NICE NG128 rec 1.1.2 [2008] says to exclude hypoglycaemia, so glucose context belongs here. | G08 |
| R12 | Systemic cancer therapy planning | Cardiac history or LVEF found | lvef; hypertension_hx; coronary_hf | Show heart-function facts with dates. | G11 (cardio-oncology to source) |
| R13 | Radiotherapy planning | Pacemaker or ICD found | cardiac_device | Show the device and date; a device is never hidden for this task. | G17 (to source) |
| R14 | Weight or diabetes review | Two or more dated weights | weight; diagnoses_all; med_antidiabetic | Compute change in code; show every input weight linked; add diagnoses and drugs as context only. | G18 |
| R15 | Follow-up review | Follow-up recommended | discharge_followups; followup_done | For each recommendation search for evidence it happened; none found is 'not documented'. | G18 |
| R16 | Medication review | Several classes found | med_current; med_stopped_changed; med_renal_relevant | Group by class using the lookup; flag same-class duplicates for the clinician. | G15 |
| R17 | Any task | Allergy documented | allergies | Allergies are P1 in every bundle that can lead to a prescription or procedure; never hidden by detail level. | G13 |
| R18 | Surgery or anaesthesia | Reduced or unknown EF, or heart failure | lvef; ecg_rhythm; coronary_hf | Show EF with date and wording; NICE NG45 section 1.9 covers resting echocardiography. | G01,G07 |

| ID | When the task is | And this fact is found | Surface | How to display | Sources |
|---|---|---|---|---|---|
| R19 | Surgery or procedure | Carotid stenosis found | carotid_stenosis; stroke_tia_hx | Show side, degree and symptoms. | G07,G08 |
| R20 | Minor or dental procedure | Valve disease found | valves; med_antithrombotic | Show valve status and blood-thinner status. | G18 |
| R21 | Any task | Newest value older than 12 months before the anchor | (all latest_value slots) | Show 'as of <month year>' on the value (source dates are month precision); never present an old value as current. | code only |
| R22 | Any task | Any medication is shown | (the shown medication slot) + label panel | Show the label panel for the ingredient (section 11): listed contraindications beside the related patient facts (found, documented as none, or not documented). Never state that a drug is contraindicated for this patient. | L10 (production), L02-L09 (prototype) |

# 8 Deterministic consistency checks

Code-only checks that produce contradiction candidates. Every card links all passages involved. Guards exist because a false alarm cancels a true find. Global guards: ignore small measurement differences, values that changed over time, and differences from the JSON extraction.

| ID | Check | Slots | Method | False-alarm guard |
|---|---|---|---|---|
| X01 | BMI recomputation | height_bmi; weight | BMI = kg / m^2 using height and weight from the same letter. | Tolerance 0.5 kg/m^2 for rounding. Different letters are not compared. |
| X02 | Age against date of birth | age_sex | Compute age at the document date (and at the anchor date) from the date of birth; compare with any stated age. Never use the age field in patient.json (it is today's age). | Only flag if the gap is 1 year or more. |
| X03 | Laterality conflict | carotid_stenosis; tumour_site_laterality; stroke_tia_hx; eye_history | Extract side tokens (links, rechts, beidseits, left, right) per finding; compare for the same finding. | Same finding and same time frame only; 'both' vs one side is shown as a note, not an alarm. |
| X04 | Event date conflict | hospital_stays; procedures_all; planned_procedure; prior_similar_episode | Same event type and entity with two different dates. | Month-year is compatible with a full date in that month; planned vs actual are different events. |
| X05 | Dose conflict | med_current and class slots | Same drug with different dose in documents from the same time window. | Dose changes across time are not contradictions; compare only within 7 days or the same letter. |
| X06 | Allergy status conflict | allergies | 'keine bekannten Allergien' next to a listed allergen in the same time window. | Allergy later documented is a change over time, not a conflict. |
| X07 | HbA1c unit consistency | hba1c | IFCC mmol/mol = (NGSP % - 2.15) x 10.929; compare when both units appear for one test. | Flag only if the converted values differ by 2 mmol/mol or more. |
| X08 | T category against tumour size | tnm_grade; tumour_site_laterality | Breast: T1 up to 20 mm, T2 over 20 up to 50 mm, T3 over 50 mm. | Same pathology report only; clinical vs pathological size differences are false alarms. |
| X09 | EF value against wording | lvef | Numeric EF against words such as normal or reduced. | Gross mismatches only; thresholds vary between sources. |
| X10 | Negated vs affirmed diagnosis | diagnoses_all | 'kein / ohne X' beside 'X' for the same time window. | Resolved or historical diagnoses ('Z.n.') are changes over time. |

# 9 Guideline source register and source policy

## Source policy

| Tier | Use | Rule |
|---|---|---|
| A: primary guidelines | Cited on screen | Austrian societies first (ÖGARI, ÖDG and similar), then German and European, then NICE labelled as a UK guideline. Cite short name, ID or section, year and link. Write the 'why' in the team's own words. |
| B: study notes and question banks (for example PassMedicine) | Discovery only | Use to find which guideline sits behind a rule, then cite the guideline. Do not cite or copy the notes; their reuse terms were not checked in this session, so treat them as copyrighted. |
| C: anything unverified | Not shown | Entries marked M below stay off the screen until verified. |

## Licensing note (not legal advice)

NICE publishes a UK Open Content Licence that allows free reuse in a UK setting only; use outside the UK needs a NICE international licence. NICE also asks for a specific attribution statement, and BNF/BNFC content is excluded from its licence. For a prototype, a short reference with link and a rationale in your own words is the low-risk route; check with NICE before any real deployment.

## Register

Each link opens the guideline page or DOI. Years matter: the demo documents are from 2015-2016, so the 'Era' column says whether the cited text could have been in force then.

Suggested on-screen citation: Body ID (year) - recommendation number - link. Example: NICE NG28 (2015, updated 2026) - 1.6.1 - nice.org.uk/guidance/ng28.

| ID | Title and citation | Body and place | Era vs April 2016 | What it supports (points seen) | Mark | Link |
|---|---|---|---|---|---|---|
| G01 | **Routine preoperative tests for elective surgery (NG45)**<br>Published 5 Apr 2016; last updated Jul 2024 | NICE, UK (England) | Contemporary: published the same month as the case (text since updated) | Ties tests to comorbidity. HbA1c section 1.6 (offer to people with diabetes not tested in the last 3 months); echocardiography section 1.9; also kidney function, haemostasis and ECG.<br>*Note: Excludes cardiothoracic surgery, neurosurgery and pregnancy. Check the surgery-grade and ASA tables.* | V | www.nice.org.uk/guidance/ng45 |
| G02 | **Präoperative Evaluation erwachsener Patienten vor elektiven, nicht herz-thoraxchirurgischen Eingriffen. Gemeinsame Empfehlung DGAI, DGCH, DGIM**<br>2017 edition (DOI 10.1007/s00101-017-0321-5); revision published 3 May 2024 (DOI 10.1007/s00101-024-01408-2) | DGAI / DGCH / DGIM, Germany (DACH) | 2017 edition is closest to 2016; 2024 is current | Part A general principles, Part B extended diagnostics, Part C handling of long-term medication (feeds the antithrombotic, antidiabetic and ARB rules).<br>*Note: German societies: check whether an Austrian society endorses it before calling it Austrian.* | V | doi.org/10.1007/s00101-024-01408-2 |
| G03 | **Präoperative Blutungsanamnese (Pfanner et al.)**<br>Anaesthesist 2007;56:604-611 | ÖGARI perioperative coagulation working group, Austria | Older than the case; check for an update | Recommends a standardised bleeding-history questionnaire; patients on coagulation-relevant drugs need laboratory tests, starting with platelet count (supports bleeding-history, coagulation and blood-count slots).<br>*Note: The excerpt seen was cut off after the platelet count; read the paper for the full parameter list.* | V | doi.org/10.1007/s00101-007-1182-0 |
| G04 | **ÖGARI working-group guideline on pre-operative evaluation (two-category risk scheme: minor vs major surgery)**<br>Year not found | ÖGARI, Austria | Unknown | Described in a Swiss guideline (IHAMZ) as an Austrian working-group guideline; the primary ÖGARI document was not located.<br>*Note: Do not show on screen until the primary source is found. The link below is the Swiss secondary source.* | M | www.guidelines.fmh.ch/downloads/23084185/download-de.pdf |

| ID | Title and citation | Body and place | Era vs April 2016 | What it supports (points seen) | Mark | Link |
|---|---|---|---|---|---|---|
| G05 | **Diabetes mellitus - Leitlinien für die Praxis (ÖDG)** Wien Klin Wochenschr; editions seen: 2007, 2009, 2012, 2019, 2023 (vol 135, Suppl 1); 2026 pocket guide | ÖDG, Austria | Edition in force in April 2016 not yet identified (the 2012 foreword announced the next edition for 2015) | Diabetes review: drugs, targets, complications. The 2026 pocket guide states it summarises the sixth full revision. *Note: Locate the 2015/2016 edition for a period-correct citation; the 2023 text is current.* | V | www.oedg.at/pdf/OEDG-Leitlinien-2023.pdf |
| G06 | **Pre-operative evaluation of adults undergoing elective noncardiac surgery: updated guideline from the ESA (De Hert et al.); 2025 ESAIC update** Eur J Anaesthesiol 2018;35(6):407-465 (literature search to May 2016); update 2025;42(1):1-35 | ESA / ESAIC, Europe | 2018 is contemporary; a 2025 update exists | General pre-operative assessment framework. The 2025 update gives 55 recommendations agreed by Delphi vote. *Note: Choose one edition and label it with its year. 2025 update DOI: 10.1097/EJA.0000000000002069.* | V | doi.org/10.1097/EJA.0000000000000817 |
| G07 | **ESC/ESA guidelines on non-cardiac surgery: cardiovascular assessment and management (2014); ESC guidelines on non-cardiac surgery (2022)** Eur Heart J 2014;35(35):2383-2431; Eur Heart J 2022;43(39):3826-3924 | ESC / ESA (2022 endorsed by ESAIC), Europe | 2014 is contemporary; 2022 is later | 2014 text highlights functional capacity and is the first to recommend peri-operative regimens for dual antiplatelet therapy and new oral anticoagulants. 2022 key messages stress patient-reported functional capacity and antithrombotic management. *Note: Use 2014 for period-correct citations and 2022 for current practice.* | V | doi.org/10.1093/eurheartj/ehu282 |
| G08 | **Stroke and transient ischaemic attack in over 16s: diagnosis and initial management (NG128)** Published 1 May 2019 (replaced CG68, 2008) | NICE, UK (England) | Later than the case; recommendations tagged [2008] were in force in 2016 | Rec 1.1.2: exclude hypoglycaemia as the cause of sudden-onset neurological symptoms [2008]. Rec 1.1.4: aspirin 300 mg for suspected TIA [2019]. Includes imaging for suspected TIA. *Note: Cite recommendation numbers with their [year] tags.* | V | www.nice.org.uk/guidance/ng128 |
| G09 | **Hypertension in adults: diagnosis and management (NG136)** Published 28 Aug 2019; last updated 21 Nov 2023 (replaced CG127, 24 Aug 2011) | NICE, UK (England) | Later than the case; CG127 was in force in 2016 | Blood pressure measurement, diagnosis and treatment; also replaces the blood pressure section of NG28. *Note: A 2024 ESC hypertension guideline also exists; choose one and label it.* | V | www.nice.org.uk/guidance/ng136 |
| G10 | **Type 2 diabetes in adults: management (NG28)** Published 2 Dec 2015; last updated Feb 2026 (replaced CG87) | NICE, UK (England) | Contemporary in its 2015 form; the text has since been revised | Section 1.6 blood glucose management including individual HbA1c targets; rec 1.5.1: do not offer aspirin or clopidogrel to adults with type 2 diabetes without cardiovascular disease [2015]. *Note: Check which recommendations changed after 2015 before citing a section.* | V | www.nice.org.uk/guidance/ng28 |
| G11 | **Early and locally advanced breast cancer: diagnosis and management (NG101)** Published 18 Jul 2018; last updated 16 Jan 2024 | NICE, UK (England) | Later than the case; carries 2009 recommendations that were in force in 2016 | Rec 1.3.1: assess ER, PR and HER2 together at initial histopathological diagnosis [2018]. Rec 1.2.1: pre-treatment ultrasound of the axilla [2009]. Sections on surgery, radiotherapy and systemic therapy planning. *Note: Cite recommendation numbers with their [year] tags.* | V | www.nice.org.uk/guidance/ng101 |
| G12 | **Acute kidney injury: prevention, detection and management (NG148)** Published 18 Dec 2019; last updated 16 Oct 2024 (replaced CG169, 2013) | NICE, UK (England) | Later than the case; CG169 was in force in 2016 | 2024 contrast recommendations: do not delay iodine-based contrast in a clinically significant emergency; before non-urgent contrast CT assess for existing kidney disease (1.1.8); if screening is negative and the person is stable, proceeding without further blood tests may be considered (1.1.9). *Note: This softens how hard the contrast bundle should push kidney tests: show kidney function if documented, never imply contrast must wait for it.* | V | www.nice.org.uk/guidance/ng148 |

| ID | Title and citation | Body and place | Era vs April 2016 | What it supports (points seen) | Mark | Link |
|---|---|---|---|---|---|---|
| G13 | **Drug allergy: diagnosis and management (CG183)**<br>Published 3 Sep 2014 | NICE, UK (England) | Contemporary | Names poor documentation of drug allergy as a major issue and aims to ensure allergies are recorded properly (supports the allergy slot and the split between 'documented as none' and 'not documented').<br>*Note: Recommendation numbers were not read; cite the guideline, not a numbered recommendation, until checked.* | V | www.nice.org.uk/guidance/cg183 |
| G14 | **Contrast media safety: ESUR guidelines (contrast nephropathy 2011; post-contrast AKI recommendations 2018, in guideline version 10)**<br>Eur Radiol 2011;21(12):2527-2541; Part 2 recommendations 2018 | ESUR Contrast Media Safety Committee, Europe | 2011 is contemporary; 2018 is later | 2011: patients with eGFR 60 or above can continue metformin. 2018: stop metformin from the time of contrast if eGFR is below 30. Also covers risk factors, hydration and dialysis.<br>*Note: Verify exact thresholds in the full text before any on-screen wording; the app shows facts, not threshold advice.* | V | doi.org/10.1007/s00330-011-2225-0 |
| G15 | **Medicines optimisation: the safe and effective use of medicines to enable the best possible outcomes (NG5)**<br>Published 4 Mar 2015 | NICE, UK (England) | Contemporary | Covers medicines reconciliation, medication review and decision aids for people taking one or more medicines.<br>*Note: UK guideline: label it as such.* | V | www.nice.org.uk/guidance/ng5 |
| G16 | **Perioperative care in adults (NG180)**<br>Published 19 Aug 2020 | NICE, UK (England) | Later than the case | Covers all phases of perioperative care for adults having elective or emergency surgery, including dental surgery.<br>*Note: Supplementary for bundles B01 and B15.* | V | www.nice.org.uk/guidance/ng180 |
| G17 | **Austrian society guidelines still to locate: stroke/TIA, general hypertension, breast cancer treatment**<br>Not located | Austrian societies, Austria | Unknown | Nothing confirmed yet.<br>*Note: Search the Austrian stroke, hypertension and senology societies. Do not cite until a document is found.* | M | none yet |
| G18 | **No clinical guideline: workflow or documentation purpose; cite local discharge or handover policy**<br>n/a | Local policy, Local | n/a | Bundles about information completeness rather than a clinical decision.<br>*Note: State this honestly on screen: 'workflow rationale, no guideline'.* | n/a | none yet |
| G19 | **Individualisierung der antihypertensiven Therapie bei Patienten mit Diabetes mellitus (ÖDG guideline)**<br>Wien Klin Wochenschr 2019;131(Suppl 1):S124-S135; update 2023 | ÖDG, Austria | Later than the case | Austrian guideline on blood pressure treatment in diabetes (ACE inhibitors, or alternatively ARBs, and combination therapy). Supports the blood pressure bundle for patients with diabetes.<br>*Note: Read the 2023 update before citing any target values.* | V | doi.org/10.1007/s00508-019-1460-2 |
| G20 | **Empfehlungen der Österreichischen Gesellschaft für Senologie: Nachsorge nach Mammakarzinom**<br>Senologie 2020;17(04):232-235 | Österreichische Gesellschaft für Senologie, Austria | Later than the case | Austrian follow-up recommendations after breast cancer. Supports follow-up slots, not surgical planning.<br>*Note: Seen in a citation-index record only; open the journal page before citing.* | V | ouci.dntb.gov.ua/works/4YObnX39 |
| G21 | **S3-Leitlinie Früherkennung, Diagnostik, Therapie und Nachsorge des Mammakarzinoms (AWMF register 032-045OL)**<br>Consultation version 0.4.1 (Aug 2017) seen in citations; later versions exist | German guideline programme (AWMF), Germany (DACH) | Later than the case | German-language breast cancer guideline used alongside German and international recommendations; candidate companion to NG101 for surgical-planning slots.<br>*Note: Existence confirmed from citations only; fetch the current version from AWMF and confirm numbering.* | V | none yet |

# 10 Validation

## Demo coverage: each of the 15 demo questions must map to a P1 slot of the right bundle

| Question | Situation | Slot(s) | Where |
|---|---|---|---|
| 1.1 | Pre-operative anaesthesia check | med_antithrombotic | B01 P1 |
| 1.2 | Pre-operative anaesthesia check | allergies | B01 P1 |
| 1.3 | Pre-operative anaesthesia check | lvef | B01 P1 |
| 2.1 | Surgery planning after tumour board | cancer_histology; tnm_grade; tumour_site_laterality | B02 P1 |
| 2.2 | Surgery planning after tumour board | receptors_ki67 | B02 P1 |
| 2.3 | Surgery planning after tumour board | board_recommendation; planned_procedure | B02 P1 |
| 3.1 | Diabetes review at the GP | med_antidiabetic | B03 P1 |
| 3.2 | Diabetes review at the GP | hba1c | B03 P1 |
| 3.3 | Diabetes review at the GP | weight | B03 P1 (+ rule R14) |
| 4.1 | Emergency department at night | prior_similar_episode; stroke_tia_hx; carotid_stenosis | B04 P1 |
| 4.2 | Emergency department at night | allergies; vascular_imaging; med_antidiabetic; kidney_function | B05 P1, implied by B04 |
| 4.3 | Emergency department at night | med_antihypertensive | B04 P1 |
| 5.1 | Open follow-ups from the 2015 stay | discharge_followups | B06 P1 |
| 5.2 | Open follow-ups from the 2015 stay | followup_done | B06 P1 (absence = not documented) |
| 5.3 | Open follow-ups from the 2015 stay | carotid_stenosis | B06 P1 |

## Extra test situations (hold-out set; keep out of the knowledge base)

| # | Role | Situation text | Expected routing |
|---|---|---|---|
| 1 | Anaesthetist | Sedation for a colonoscopy next week | B01 |
| 2 | Orthopaedic surgeon | Knee replacement pre-assessment | B01 + surgeon modifiers |
| 3 | Radiographer | CT angiogram request, safety check | B05 |
| 4 | Pharmacist | Medication reconciliation on admission | B07 |
| 5 | Cardiologist | Outpatient review of blood pressure and heart function | B09 + cardiologist modifiers |
| 6 | Neurologist | TIA clinic referral | B04 + B17 |
| 7 | Ophthalmologist | Transient vision loss, left eye | B14 |
| 8 | Dentist | Tooth extraction under local anaesthetic | B15 |
| 9 | Paramedic | Collapse at home, handover to the emergency department | B16 |
| 10 | Dietitian | Unintentional weight loss | B10 |
| 11 | Physiotherapist | Shoulder mobility after breast surgery | B13 |
| 12 | Medical oncologist | Adjuvant therapy discussion | B11 |
| 13 | Radiation oncologist | IORT planning | B12 |

| # | Role | Situation text | Expected routing |
|---|---|---|---|
| 14 | Ward nurse | Nursing admission assessment | B08 |
| 15 | GP | Annual hypertension review | B09 |
| 16 | GP | Which follow-ups are overdue? | B06 |
| 17 | Diabetologist | Peri-operative diabetes plan before surgery | B03 + B01 (multi-bundle test) |
| 18 | Vascular surgeon | Carotid endarterectomy consideration | B17 |
| 19 | Anästhesist (German wording) | Prämedikationsvisite vor Quadrantektomie | B01 (German alias test) |
| 20 | Emergency physician | Chest pain at night | No bundle: should fall back to B18 with a visible note |
| 21 | Admitting doctor | Medical admission with confusion | No bundle: fallback B18 |
| 22 | Locum GP | New to the practice, give me a summary | B18 by design |

# 11 Medication label reference (contraindications beside patient facts)

Whenever a medication is shown, the screen can add a collapsed label panel for its ingredient. The source is the official product information (SmPC), not a clinical guideline: guidelines such as NICE NG148 and ESUR (section 9) cover the contrast and kidney protocols, while the label lists the formal contraindications. The panel shows label statements beside patient facts and never concludes anything about the patient.

## Display rules

| Rule | Detail |
|---|---|
| Source of truth | Official product information (SmPC, section 4.3 Contraindications; section 4.4 warnings). In production use the Austrian Fachinformation from the BASG register (L10). The prototype uses EU or UK labels for the same ingredient (L01-L09). |
| Match by ingredient | Match the shown medication to a label through the generic name or ATC code, never the brand string. Brand names in the notes are only search aids. |
| Two lists | Contraindications (section 4.3) are shown as 'listed contraindications'. Warnings (section 4.4 or 4.2) are shown separately and only in long mode or when the task matches (for example contrast imaging). |
| Beside patient facts | Each listed item is shown with the related patient facts: found (with source link), documented as none, or not documented. The screen never says a drug is contraindicated for this patient. |
| Short mode | P1 items only, and only where the related slot is found or not documented. The panel is collapsed with a count badge. |
| Long mode | All items, warnings, and population items (pregnancy, children). Population items are hidden when the age rule excludes them. |
| Class checks | Interaction contraindications (for example candesartan with aliskiren in diabetes, aspirin with methotrexate at 15 mg per week or more) run as code over the medication list and diabetes or kidney facts. |
| Show the label | Every panel names the label used, its authority and date, and links to it. Era note: label text is current, while the 2015-2016 prescriber used the label of that time. |
| Labels differ | Different products of one ingredient can list different contraindications (for example some aspirin labels also list concurrent oral anticoagulation). Never merge labels silently. |
| Not used | BNF and BNFC content (L12). Commercial drug databases unless licensed. |

## Label sources used by the prototype

| ID | Document | Authority | Version seen | Used for | Link | Note |
|---|---|---|---|---|---|---|
| L01 | Glucophage 500 mg / 850 mg film-coated tablets: SmPC | Merck, published on EMC (UK) | Updated 22 Dec 2025 | Metformin warnings: contrast and surgery wording (section 4.4) | www.medicines.org.uk/emc/product/7759/smpc/print | UK label; used for the warning wording. |
| L02 | Glucophage 500 mg film-coated tablets: SmPC, licence PA2286/005/001 | HPRA (Ireland) | 1 May 2025 | Metformin section 4.3 contraindication list; surgery wording seen in the sister document PA2286/005/002 | assets.hpra.ie/products/Human/19403/Licence_PA2286-005-001_01052025155855.pdf | National label of an EU member state. |
| L03 | Actos 15 / 30 / 45 mg tablets: EMA EPAR product information, Annex I (SmPC) | EMA | Current EMA text (date not read) | Pioglitazone contraindications (section 4.3) and warnings (4.4) | www.ema.europa.eu/en/documents/product-information/actos-epar-product-information_pt.pdf | File name has the Portuguese language code but the pages seen are English; look for the _en file. |
| L04 | Pioglitazone Rowex 30 mg tablets: SmPC | HPRA (Ireland) | Printed 22 Nov 2014 | Period-matched pioglitazone list: the 2014 print shows five contraindications | assets.hpra.ie/products/Human/20332/LicenseSPC_PA0711-199-002_22112014213431.pdf | Close to the 2016 case; the current EMA text for Actos also lists uninvestigated macroscopic haematuria. |
| L05 | Byetta 5 / 10 micrograms solution for injection: EMA EPAR product information | EMA | Current EMA text (date not read) | Exenatide contraindication (section 4.3) and warnings (4.2, 4.4) | www.ema.europa.eu/en/documents/product-information/byetta-epar-product-information_en.pdf | The contraindication list is hypersensitivity only; renal, pancreatitis and gastrointestinal points are warnings. |
| L06 | Candesartan cilexetil 32 mg tablets: SmPC | Marketing-authorisation holder text on EMC (UK) | Last updated on EMC 20 Jan 2026 | Candesartan contraindications including aliskiren in diabetes or renal impairment | www.medicines.org.uk/emc/product/101839/smpc | Generic product label; use the BASG label for the Austrian brand. |

| ID | Document | Authority | Version seen | Used for | Link | Note |
|---|---|---|---|---|---|---|
| L07 | Candesartan cilexetil 4 mg tablets: SmPC | Marketing-authorisation holder text on EMC (UK) | Date not read | Pregnancy, severe hepatic impairment and cholestasis, children under 1 year (section 4.3) | www.medicines.org.uk/emc/product/4148/smpc/print | Generic product label. |
| L08 | Peneprin 75 mg / 100 mg gastro-resistant tablets: SmPC | Swedish Medical Products Agency (published label) | Date not read | Acetylsalicylic acid 100 mg contraindications | docetp.mpa.se/LMF/Peneprin%20gastro-resistant%20tablet%20ENG%20SmPC_09001bee838c7b55.pdf | Ingredient-level reference; labels differ between products. |
| L09 | Bayer Aspirin Cardio 100: product information | SAHPRA (South Africa) | 30 Jun 2022 | Corroborating list including eGFR below 30 and NYHA III-IV wording | pi-pil-repository.sahpra.org.za/wp-content/uploads/2022/08/BAYER-ASPIRIN-CARDIO-100-ENG-PI-30.06.2022.pdf | Not an EU label; corroboration only. |
| L10 | Medikamenten-Informationssystem (formerly Arzneispezialitätenregister) | BASG (Austria) | Live register | Official source of the Austrian Fachinformation, searchable by name, active substance or ATC code | www.gesundheit.gv.at/service/gesundheitssuche/medikamentensuche | The production source for the exact Austrian products; not fetched in this session. |
| L11 | EMA legal notice | EMA | Seen 29 Oct 2026 | Reuse: reproduction and citation allowed with acknowledgement; third-party content excluded | www.ema.europa.eu/en/about-us/legal-notice | Check that it covers each product-information document before copying text. |
| L12 | BNF and BNFC licensing restriction | Pharmaceutical Press | Seen 29 Oct 2026 | Why the BNF is not used | www.pharmaceuticalpress.com/?p=146126 | Says NHS bodies must not insert BNF content into third-party software and that use in AI training is prohibited. |

## Metformin

| | |
|---|---|
| **Brand in the demo cases** | Glucophage (in the demo cases) |
| **ATC** | A10BA02 (check) |
| **Drug class lookup** | biguanide |
| **Label sources** | L02, L01 |
| **Era and label notes** | Current label text. Some older or generic labels list iodinated contrast in section 4.3; the current Glucophage text puts it in section 4.4 as stop-around-imaging advice. |

| Listed contraindication (paraphrased from the label) | Kind | Priority | Patient facts shown beside it |
|---|---|---|---|
| Hypersensitivity to metformin or an excipient | hypersensitivity | P1 | Allergies and intolerances, with reaction |
| Any acute metabolic acidosis, such as lactic acidosis or ketoacidosis | acute_condition | P2 | Blood glucose, Diabetic complications |
| Diabetic pre-coma | acute_condition | P2 | Blood glucose, Diabetes: type and duration |
| Severe renal failure (GFR below 30 mL/min) | organ_function | P1 | Creatinine and eGFR |
| Acute conditions that can alter kidney function: dehydration, severe infection, shock | acute_condition | P2 | Creatinine and eGFR, Blood pressure and heart rate, Oxygen saturation, respiratory rate, temperature |
| Disease that can cause tissue hypoxia: decompensated heart failure, respiratory failure, recent myocardial infarction, shock | organ_function | P1 | Coronary disease, MI, stents, heart failure, LVEF and echo findings, Lung disease and sleep apnoea |
| Hepatic insufficiency, acute alcohol intoxication, alcoholism | organ_function | P1 | Liver function, Smoking and alcohol |

| Label warning (not a contraindication) | Shown when bundle | Patient facts shown beside it | Labels |
|---|---|---|---|
| Iodinated contrast: the label advises stopping metformin at or before the imaging procedure and not restarting for at least 48 hours, once kidney function is re-checked and stable (section 4.4). | B05 Safety check before iodinated contrast | Creatinine and eGFR, Glucose-lowering drugs, incl. insulin | L01 |
| Surgery under general, spinal or epidural anaesthesia: the label advises stopping at the time of surgery and restarting no earlier than 48 hours after surgery or resuming oral food, once kidney function is confirmed. | B01 Pre-procedure check (anaesthesia) | Creatinine and eGFR, Planned procedure and date | L02 |

## Pioglitazone

| | |
|---|---|
| **Brand in the demo cases** | Actos (in the demo cases) |
| **ATC** | A10BG03 (seen on an EMC page) |
| **Drug class lookup** | glitazone |
| **Label sources** | L03, L04 |
| **Era and label notes** | The 2014 print (L04) shows five contraindications; the current EMA text (L03) also lists uninvestigated macroscopic haematuria. |

| Listed contraindication (paraphrased from the label) | Kind | Priority | Patient facts shown beside it |
|---|---|---|---|
| Hypersensitivity to pioglitazone or an excipient | hypersensitivity | P1 | Allergies and intolerances, with reaction |
| Cardiac failure or a history of cardiac failure (NYHA I to IV) | organ_function | P1 | Coronary disease, MI, stents, heart failure, LVEF and echo findings |
| Hepatic impairment | organ_function | P1 | Liver function |
| Diabetic ketoacidosis | acute_condition | P2 | Blood glucose, Diabetic complications |
| Current bladder cancer or a history of bladder cancer | history | P1 | All diagnoses with dates and status |
| Uninvestigated macroscopic haematuria (current EMA text) | history | P2 | All diagnoses with dates and status |

| Label warning (not a contraindication) | Shown when bundle | Patient facts shown beside it | Labels |
|---|---|---|---|
| Weight gain can be a sign of heart failure, so the label asks for close weight monitoring (section 4.4). | B03 Diabetes review | Weight, LVEF and echo findings | L03 |
| Liver enzymes are to be checked before starting and periodically (section 4.4). | B03 Diabetes review | Liver function | L03, L04 |

## Exenatide

| | |
|---|---|
| **Brand in the demo cases** | Byetta (in the demo cases) |
| **ATC** | A10BJ01 (check) |
| **Drug class lookup** | glp1_agonist |
| **Label sources** | L05 |
| **Era and label notes** | Current EMA text. In the EU label the only listed contraindication is hypersensitivity; the other points are warnings. |

| Listed contraindication (paraphrased from the label) | Kind | Priority | Patient facts shown beside it |
|---|---|---|---|
| Hypersensitivity to exenatide or an excipient | hypersensitivity | P1 | Allergies and intolerances, with reaction |

| Label warning (not a contraindication) | Shown when bundle | Patient facts shown beside it | Labels |
|---|---|---|---|
| Not recommended in end-stage renal disease or severe renal impairment (creatinine clearance below 30 mL/min) (sections 4.2 and 4.4). | B03 Diabetes review | Creatinine and eGFR | L05 |
| Caution with a history of pancreatitis; stop if pancreatitis is suspected (section 4.4). | B03 Diabetes review | All diagnoses with dates and status | L05 |
| Not studied in severe gastrointestinal disease, including gastroparesis (section 4.4). | B03 Diabetes review | All diagnoses with dates and status | L05 |

## Candesartan

| | |
|---|---|
| **Brand in the demo cases** | Blopress (in the demo cases) |
| **ATC** | C09CA06 (seen on an EMC page) |
| **Drug class lookup** | arb |
| **Label sources** | L06, L07 |
| **Era and label notes** | Generic UK labels; the Austrian brand label should replace them (L10). |

| Listed contraindication (paraphrased from the label) | Kind | Priority | Patient facts shown beside it |
|---|---|---|---|
| Hypersensitivity to candesartan or an excipient | hypersensitivity | P1 | Allergies and intolerances, with reaction |
| Second and third trimesters of pregnancy | population | P3 | Age and sex |
| Severe hepatic impairment and/or cholestasis | organ_function | P1 | Liver function |
| Children below 1 year | population | P3 | Age and sex |
| Used together with aliskiren in a patient with diabetes mellitus or renal impairment (GFR below 60) | interaction | P1 | Current medication, all, Diabetes: type and duration, Creatinine and eGFR |

| Label warning (not a contraindication) | Shown when bundle | Patient facts shown beside it | Labels |
|---|---|---|---|
| Serum potassium and creatinine are to be monitored in renal impairment and during dose titration (section 4.4). | B09 Hypertension and cardiovascular review | Creatinine and eGFR, Electrolytes | L06, L07 |
| Dual blockade with ACE inhibitors, other ARBs or aliskiren is not recommended (section 4.4). | B09 Hypertension and cardiovascular review | Blood pressure drugs | L07 |

## Acetylsalicylic acid (low dose)

| | |
|---|---|
| **Brand in the demo cases** | Thrombo ASS (in the demo cases) |
| **ATC** | B01AC06 (check) |
| **Drug class lookup** | antiplatelet |
| **Label sources** | L08, L09 |
| **Era and label notes** | Labels differ between products: some also list concurrent oral anticoagulation, or asthma with nasal polyps, as contraindications. Use the label of the exact product (L10). |

| Listed contraindication (paraphrased from the label) | Kind | Priority | Patient facts shown beside it |
|---|---|---|---|
| Hypersensitivity to acetylsalicylic acid, other salicylates or NSAIDs | hypersensitivity | P1 | Allergies and intolerances, with reaction |
| Active gastric or intestinal ulcer, recurrent ulcer, or a history of gastrointestinal bleeding or perforation | history | P1 | All diagnoses with dates and status |
| Other bleeding such as cerebrovascular haemorrhage (listed in some labels) | history | P2 | Stroke, TIA, amaurosis fugax, Head imaging, CT or MRI |
| Haemorrhagic diathesis or coagulation disorders such as haemophilia and thrombocytopenia | organ_function | P1 | Coagulation tests and bleeding history, Haemoglobin and platelets |
| Severe renal impairment (eGFR below 30 in one label) | organ_function | P1 | Creatinine and eGFR |
| Severe hepatic impairment | organ_function | P1 | Liver function |
| Severe cardiac failure (NYHA III to IV in one label) | organ_function | P1 | Coronary disease, MI, stents, heart failure, LVEF and echo findings |
| Used together with methotrexate at 15 mg per week or more | interaction | P1 | Current medication, all |
| Third trimester of pregnancy (listed in some labels) | population | P3 | Age and sex |

# 12 Patient data structure notes (IBM extraction)

What the data README says and what it changes. The JSON is an index; the documents are the source of truth.

## Status vocabulary

| IBM status | Meaning | How the screen shows it |
|---|---|---|
| Diagnosis: positive | Present | Show as a fact. |
| Diagnosis: negative | Ruled out | Show as 'ruled out'; it is the 'documented as none' finding for that condition. |
| Diagnosis: suspected | Suspected | Show with the word 'suspected'; never as confirmed. |
| Diagnosis: statuspost | In the past, for example after an operation | Show as 'past' with its date. |
| Procedure: positive | Done | Show as done. |
| Procedure: statuspost | Done in the past | Show as past. |
| Procedure: recommended | Recommended, not done | Feeds follow-up and board slots; never shown as done. |

## Drug schedule decode

| Raw (morning midday evening night) | Display | Meaning |
|---|---|---|
| 1 0 0 0 | 1-0-0 | one dose in the morning |
| 0 1 0 0 | 0-1-0 | one dose at midday |
| 1 0 1 0 | 1-0-1 | one dose in the morning and one in the evening |
| 1 1 1 0 | 1-1-1 | one dose in the morning, at midday and in the evening |

## Ideas that use the structure of the data

| Idea | What to do |
|---|---|
| Use documentIds as an index | Look up which documents mention a concept, run extraction on those documents only, then validate every quote in the canonical text. Faster and cheaper; the documents stay the source of truth. |
| Route by document type | documents.json gives type, department and hospital stay. Prefer the pathology report for receptors, the tumour-board letter for the recommendation and discharge letters for follow-ups. Read the file first to confirm the real types. |
| Use the hospital stay as the unit | 'The June 2015 stay' becomes a deterministic document selection for follow-up questions. |
| Code-driven fallback for unseen roles | Map each role to ICD-10 areas and ATC groups (table below). A new role then gets a sensible screen without a handwritten bundle, which beats the general-core fallback. |
| Contradiction candidates from the JSON | The same code positive in one document and negative in another, or the same procedure with two dates. Confirm in the text only: differences from the JSON are false alarms for the metric. |
| Slot-gap audit | Run all diagnoses, procedures, drugs and measurements through the slot menu. Anything that maps to no slot is a missing slot type; add general slots, not patient-specific ones. |
| Filter junk fragments | The README says fragments such as 'zur sonographisch' appear as procedures. Drop concepts with no code and no recognisable term before using the JSON as a hint. |
| Never quote IBM's matched text | It can contain markup such as spellcorrection spans and extra spaces. Compute spans and quotes on one canonical text (entities decoded, tags stripped, whitespace collapsed). |

## Code-driven fallback: role interests by ICD-10 and ATC (codes from memory, verify)

| Role | ICD-10 areas | ATC groups |
|---|---|---|
| anaesthetist | I (circulatory), E10-E14, J40-J47, G47.3 | B01, A10, C, N02, N05 |
| cardiologist | I00-I99 | C01, C02, C03, C07, C08, C09, C10, B01 |
| diabetologist | E10-E14, N17-N19, H00-H59 | A10, C09, C10 |
| neurologist | G, I60-I69, H34.0 | B01, C10, N |
| medical_oncologist | C, D | L01, L02 |
| radiologist | N17-N19, E10-E14 | A10BA, M01A, C09 |
| ophthalmologist | H00-H59, E10-E14, I10-I15 | S01, B01 |
| vascular_surgeon | I65, I70-I79 | B01, C10 |
| pharmacist | all | all |
| gp | all | all |

# 13 Open items for clinician review

1. Every priority (P1/P2/P3) is a draft until a clinician has reviewed it; start with B01, B02, B03, B04, B05, B06 because the five demo situations depend on them.

2. Every rule in section 7 needs a clinician check and a verified guideline reference; rules surface facts and never state conclusions.

3. Guidelines marked V were located and their dates and listed points were seen in search results on 29 Oct 2026; section numbers still need a check against the full document. Entries marked M (G04, G17) are unconfirmed and stay off the screen.

4. Locate the guideline edition in force in April 2016 for each cited body, or label the citation with its own year so the screen does not imply it was current then.

5. Brand-to-generic names in section 6 are search aids only; check them against the Austrian register (L10).

6. German search hints are lexical aids for the offline extractor and a pre-filter; they must not be the only retrieval method.

7. Decide how to show 'documented as none' separately from 'not documented' in the UI.

8. Write the 22 extra test situations in section 10 as a hold-out set and keep them out of the knowledge base.

9. Section 11 (medication labels): replace the EU/UK prototype labels with the exact Austrian Fachinformation from the BASG register (L10) for each product before any real use; labels differ between products and change over time, so always show the label used and its date.

10. Section 11: a clinician must review the mapping from each listed contraindication to the patient facts shown beside it, and the P1/P3 split; the panel is label information, not a clinical decision.

11. Confirm the ATC codes marked 'check' and the ICD-10 prefixes in section 5 against the official code lists before relying on them for routing.

12. Read documents.json to confirm the real document types before building document-type routing (section 12).